

Experiment No.	1
Experiment Title	Working with Python Packages: NumPy, SciPy, Scikit-Learn, Matplotlib, and Pandas (Loan Prediction Dataset)
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Register Number	3122247001063
Faculty	Dr. Poreddy Ajay Kumar Reddy
Submission Date	August 2, 2026

1 Objective

The core goals of this laboratory exercise comprise:

- Master fundamental data manipulation, array operations, statistical routines, and data visualization primitives across the standard Python scientific stack (NumPy, SciPy, Scikit-Learn, Matplotlib, Pandas).
- Construct a reusable, automated Exploratory Data Analysis (EDA) engine capable of ingesting arbitrary tabular data and extracting statistical parameters, missing value profiles, and distribution visual shapes.
- Categorize five standard benchmark datasets (Iris, Loan Amount Prediction, Diabetes, Email Spam, MNIST) by ML paradigm, target variable characteristics, feature selection methodologies, and model choices.
- Perform full automated EDA on the continuous **Loan Amount Prediction Dataset**, analyzing generated visual outputs and statistical summaries.

2 Problem Statement

Prior to modeling real-world tabular data, data practitioners must perform comprehensive data auditing to identify skewness, unexpected multi-modal distributions, correlation structures, and missing entries. Manual analysis for every newly ingested dataset introduces code redundancy and inefficiency. This experiment addresses the challenge by implementing an end-to-end automated EDA tool using Python's numerical stack to load datasets, generate statistical summaries, and save informative visual representations. Furthermore, standard benchmark datasets are mapped into supervised and unsupervised paradigms to structure future model selection.

3 Dataset Description

The empirical analysis in this section focuses on the ****Loan Amount Prediction Dataset****.

Dataset Name	Loan Amount Prediction Dataset
Dataset Source	UCI Machine Learning Repository / Kaggle
Number of Samples	614
Number of Features	13 (5 continuous/discrete numerical, 8 categorical)

Number of Classes	N/A (Continuous Target: LoanAmount)
Missing Values	Present (Gender, Married, Dependents, Self_Employed, LoanAmount, Loan_Amount_Term, Credit_History)
Train-Test Split	N/A (Exploratory Analysis Phase)

The dataset consists of the following attributes:

- **Loan_ID**: Unique identifier string for each loan application.
- **Gender**: Applicant gender (Male / Female).
- **Married**: Marital status of the applicant (Yes / No).
- **Dependents**: Count of financial dependents (0, 1, 2, 3+).
- **Education**: Educational level (Graduate / Not Graduate).
- **Self_Employed**: Employment type indicator (Yes / No).
- **ApplicantIncome**: Primary applicant monthly income in USD.
- **CoapplicantIncome**: Additional co-applicant monthly income in USD.
- **LoanAmount**: Requested credit amount in thousands of USD (Target).
- **Loan_Amount_Term**: Loan repayment period expressed in months.
- **Credit_History**: Credit guideline compliance indicator (1.0 = positive, 0.0 = negative).
- **Property_Area**: Geographical location of property (Urban / Semiurban / Rural).
- **Loan_Status**: Historical loan approval status (Y / N).

4 Exploratory Data Analysis

The automated EDA framework generated 28 diagnostic graphics saved in the `figures_loan/` directory. Individual figures and corresponding analytical inferences are presented below.

4.1 Histograms of Numerical Features

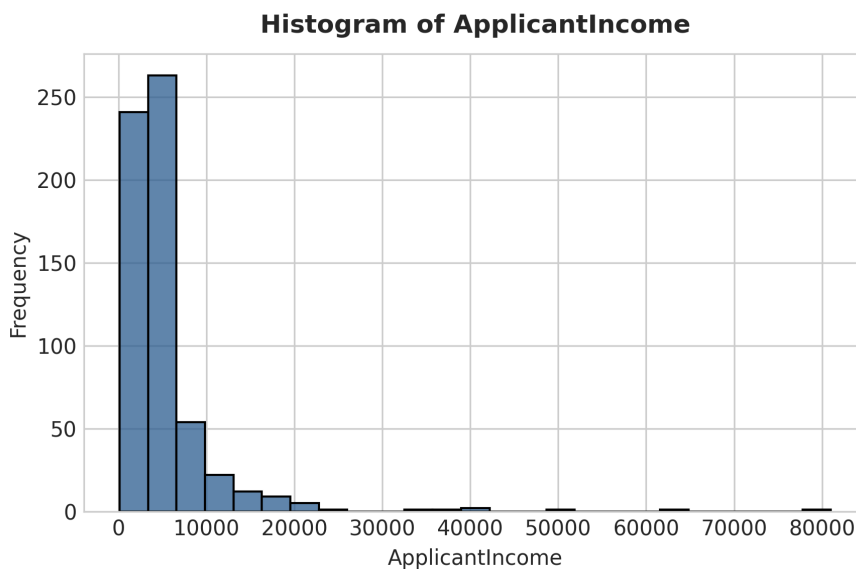


Figure 1: Histogram of Applicant Income

Inference: The applicant income distribution exhibits pronounced right skewness. Most applicants report monthly incomes below \$10,000, while a thin tail reaches toward \$80,000. Applying a logarithmic transformation prior to model fitting is advisable to mitigate variance instability.

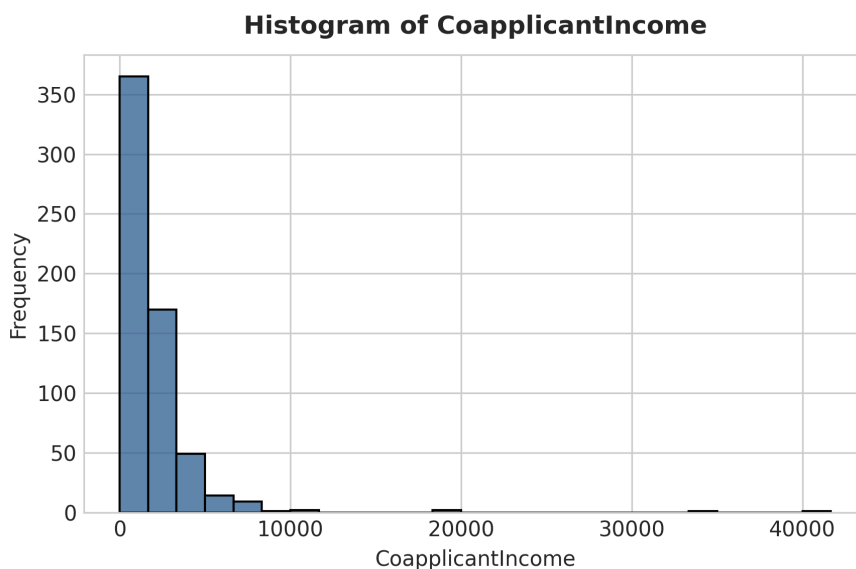


Figure 2: Histogram of Co-applicant Income

Inference: Co-applicant income features a massive spike at \$0, representing applications submitted without a co-borrower. Non-zero entries show positive skewness stretching up to \$40,000.

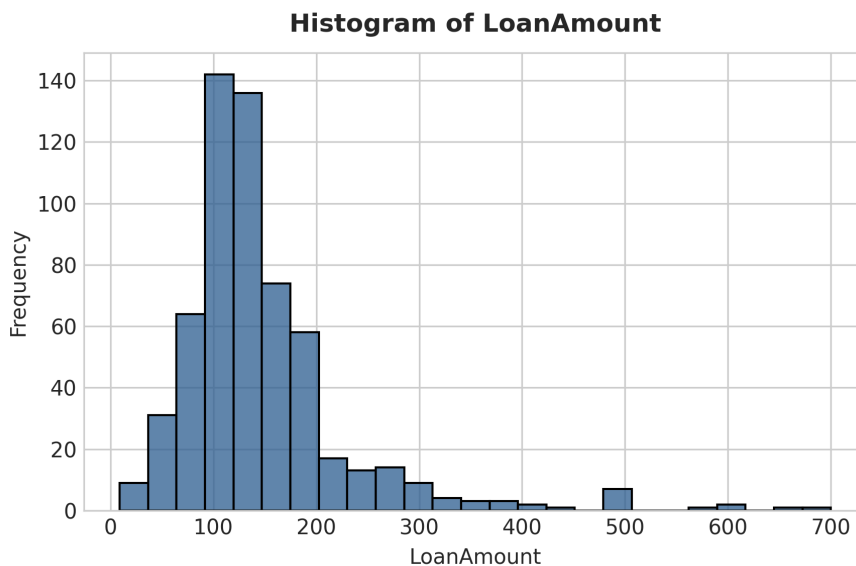


Figure 3: Histogram of Loan Amount

Inference: The target variable (`LoanAmount`) displays a unimodal distribution centered around \$130,000-\$150,000, with minor right-skewness and values spanning \$9,000 to \$700,000.

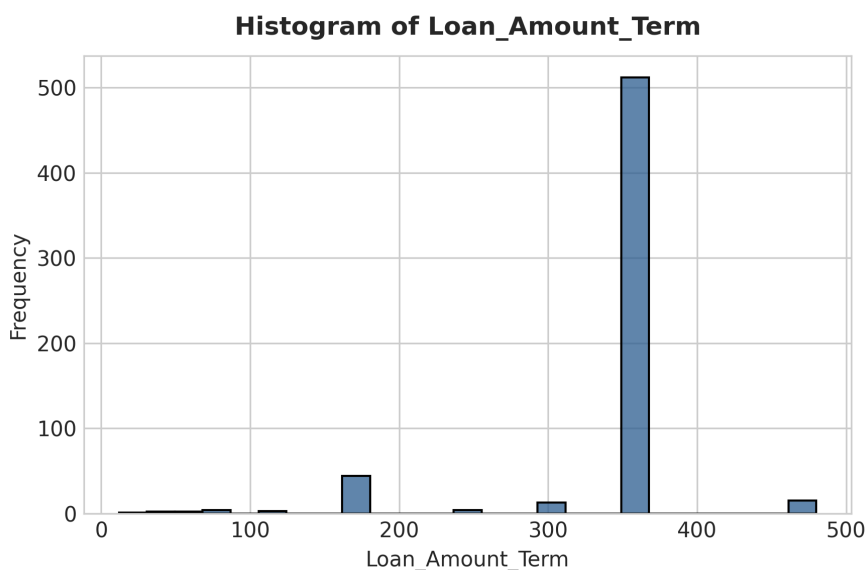


Figure 4: Histogram of Loan Amount Term

Inference: The repayment term distribution is discrete, dominated heavily by the standard 360-month (30-year) mortgage period. Intermediate terms like 180 or 240 months appear far less frequently.

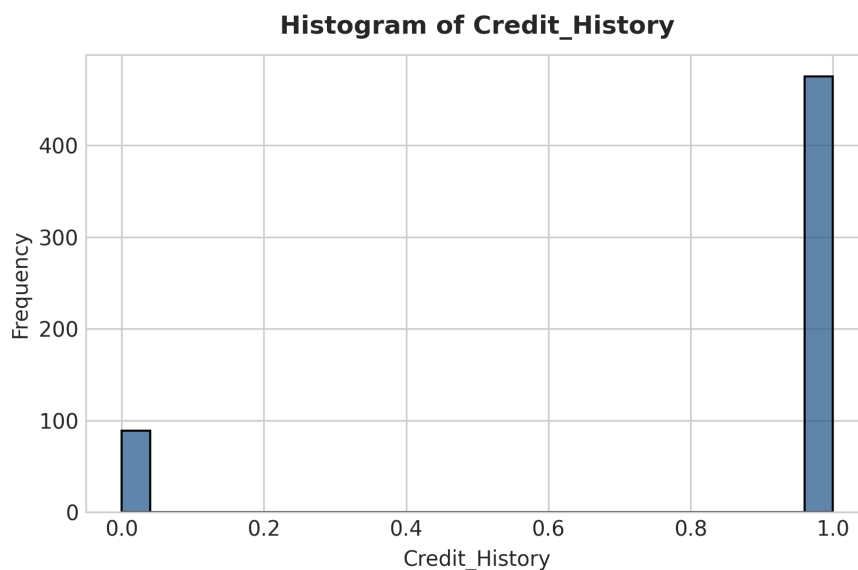


Figure 5: Histogram of Credit History

Inference: Credit history records fall into two binary bins. A significant majority of applicants possess a positive history (`Credit_History` = 1.0), indicating potential class imbalance in this attribute.

4.2 Box Plots of Numerical Features

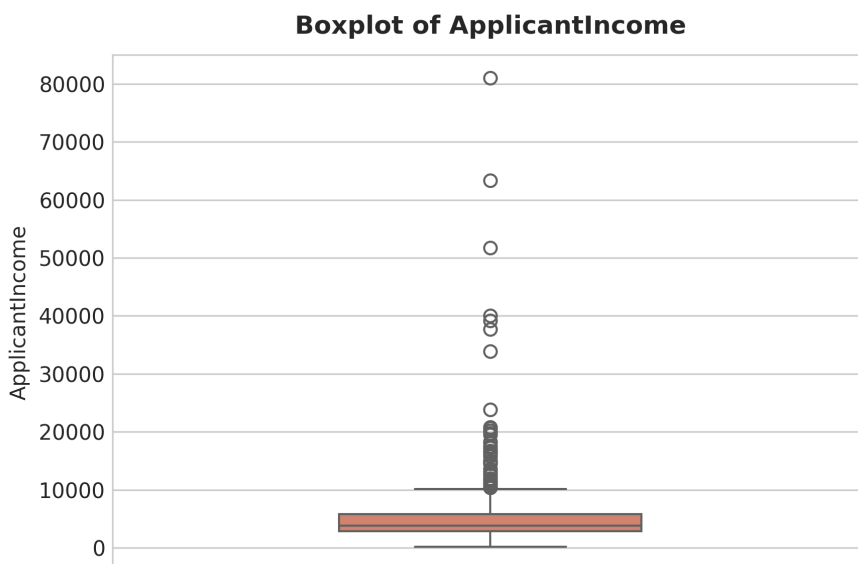


Figure 6: Box Plot of Applicant Income

Inference: The boxplot displays a compact interquartile range (IQR) with median income around \$3,812. Numerous upper outliers extend beyond the whisker bound, confirming high income dispersion.

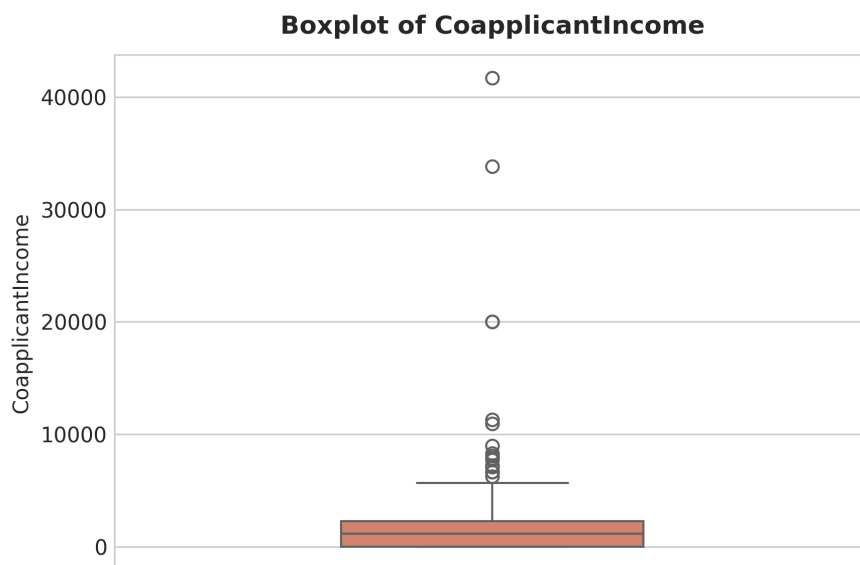


Figure 7: Box Plot of Co-applicant Income

Inference: Co-applicant income shows a low median near \$1,188, alongside multiple extreme values exceeding \$10,000.

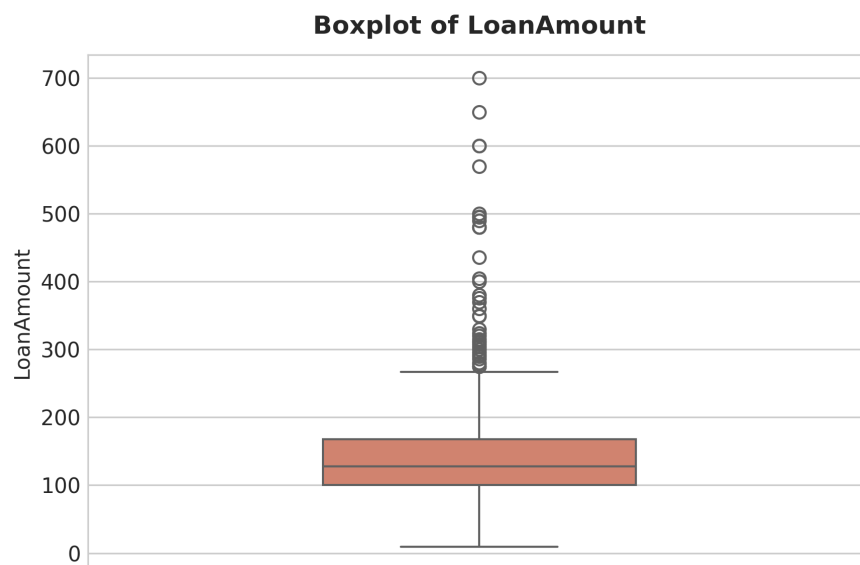


Figure 8: Box Plot of Loan Amount

Inference: The loan amount median sits near \$128,000. Outliers above \$260,000 highlight large loan applications requiring robust scaling during preprocessing.

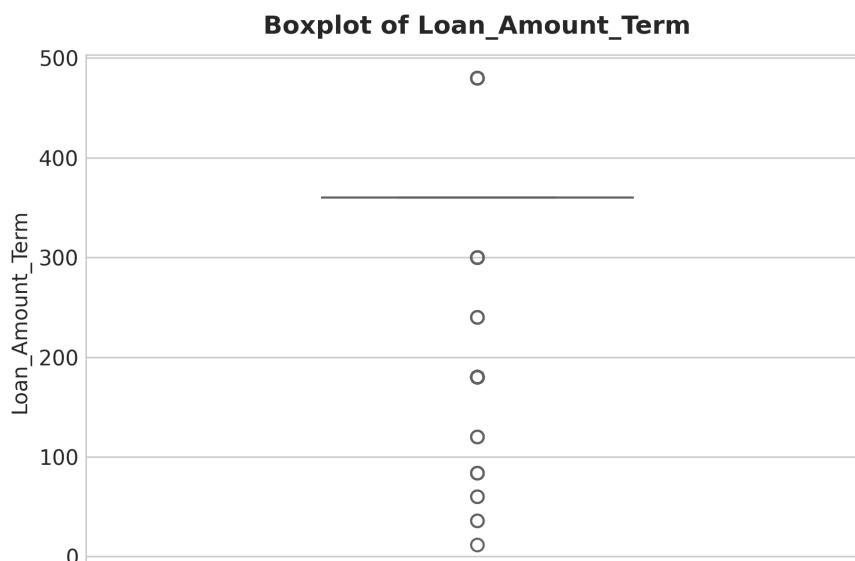


Figure 9: Box Plot of Loan Amount Term

Inference: The term variable is concentrated at 360 months. Non-standard loan durations (12 to 240 months and 480 months) appear as statistical outliers in the plot.

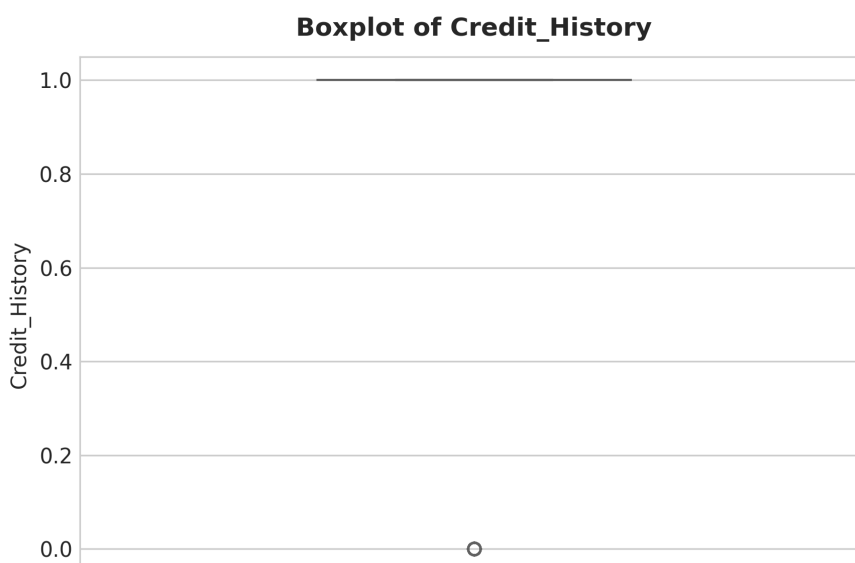


Figure 10: Box Plot of Credit History

Inference: The box plot compresses at 1.0 due to the high proportion of positive credit scores, reflecting discrete binary behavior.

4.3 Kernel Density Estimation (KDE) Plots

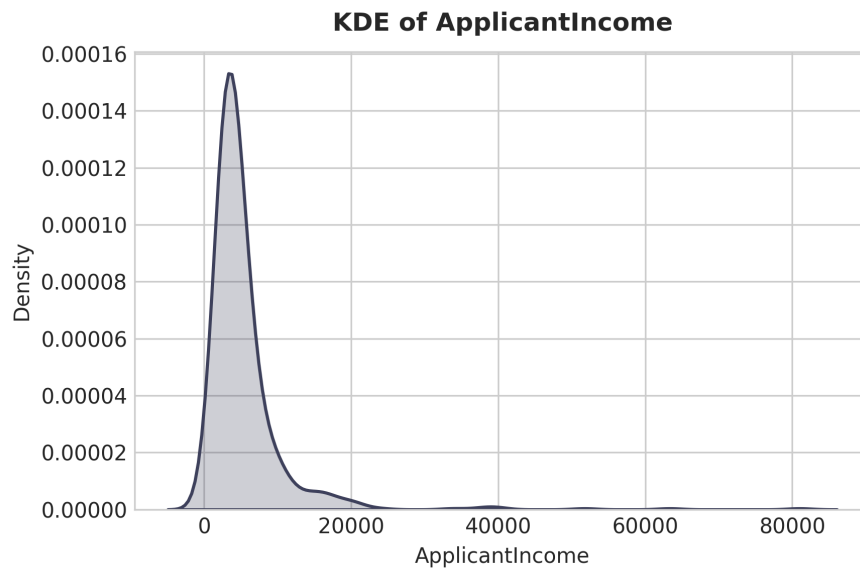


Figure 11: KDE of Applicant Income

Inference: The Gaussian smooth density curve highlights the single primary peak around \$3,500 with a long exponential decay toward higher income values.

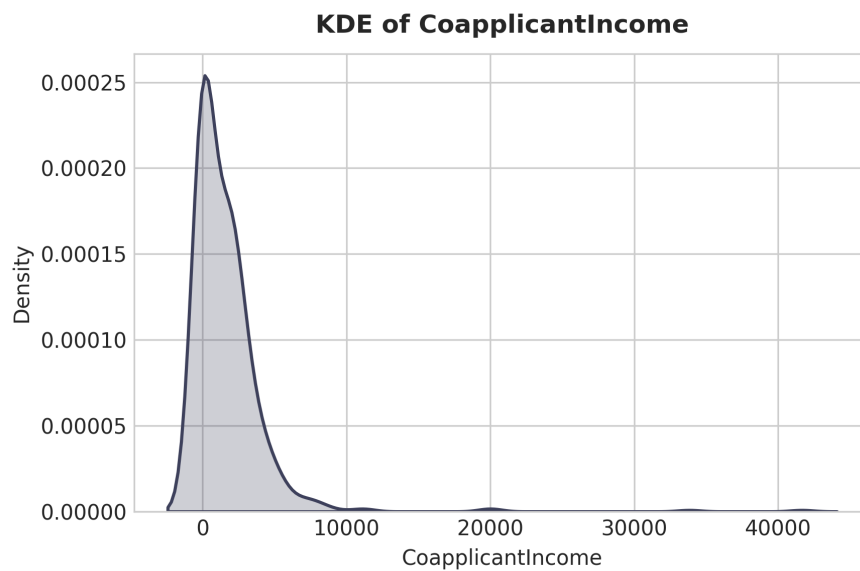


Figure 12: KDE of Co-applicant Income

Inference: Confirms a dominant density peak near \$0 representing solo applications, followed by a smoothly decaying distribution.

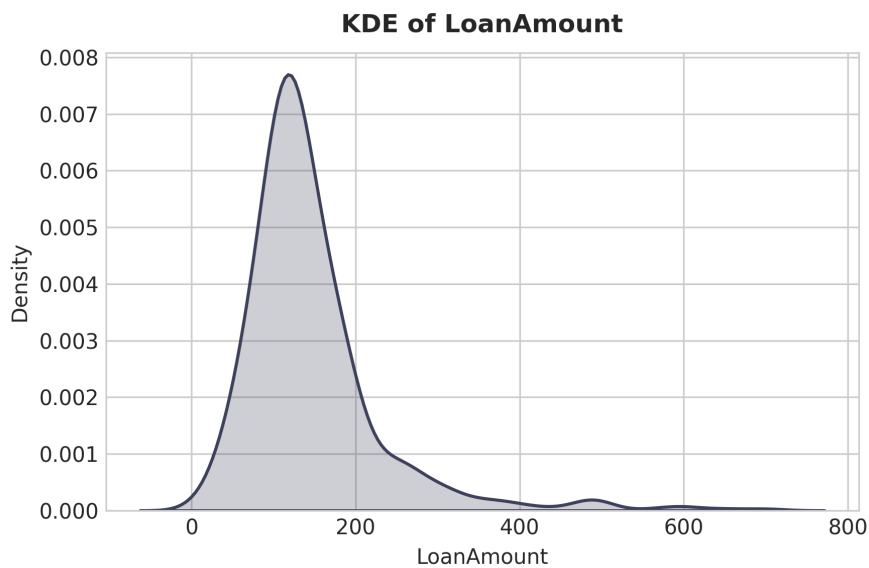


Figure 13: KDE of Loan Amount

Inference: The continuous density estimate displays near log-normal behavior, peaking cleanly around \$125,000.

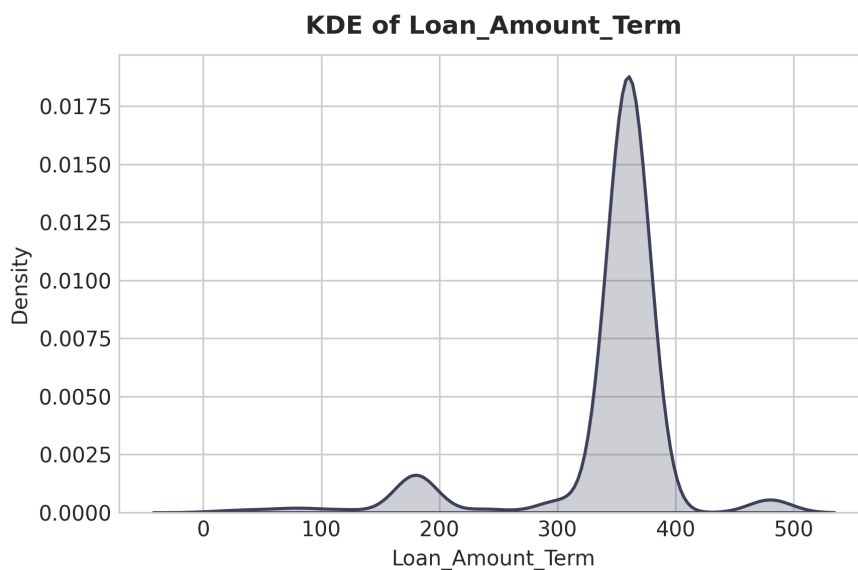


Figure 14: KDE of Loan Amount Term

Inference: Displays an extremely narrow peak centered precisely at 360 months.

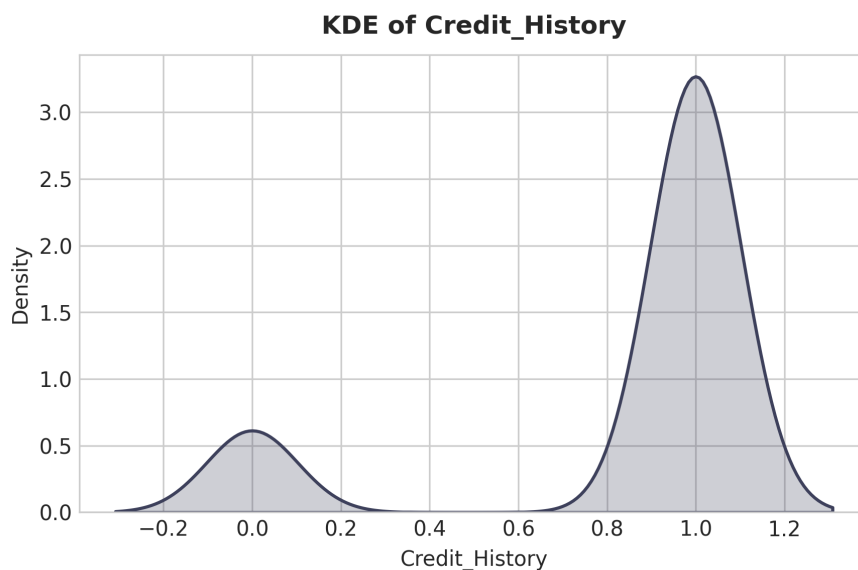


Figure 15: KDE of Credit History

Inference: Bimodal density peaks near 0.0 and 1.0 clearly illustrate the binary nature of historical credit standing.

4.4 Count Plots of Categorical Features

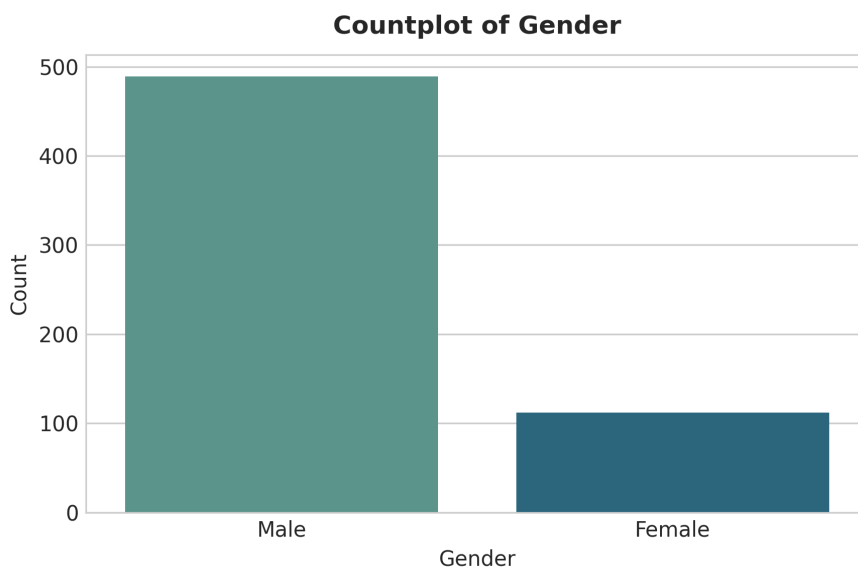


Figure 16: Count Plot of Gender

Inference: Male applicants comprise over 80% of the records, highlighting gender imbalance within the historical sampling pool.

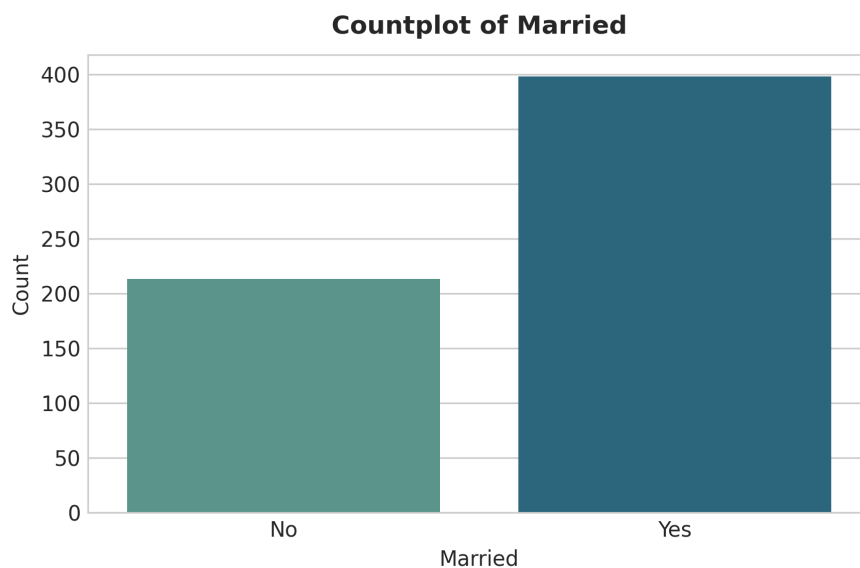


Figure 17: Count Plot of Married

Inference: Married individuals account for roughly two-thirds of total applications, reflecting joint credit applications.

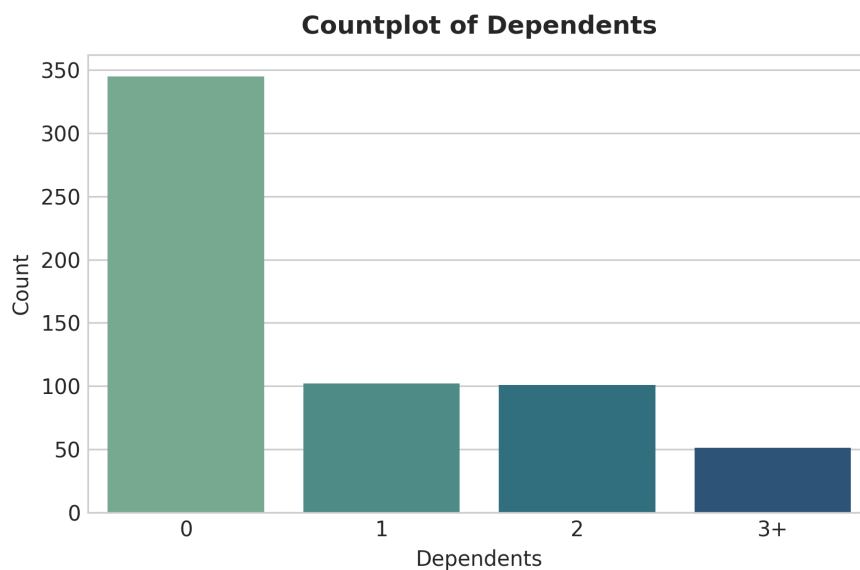


Figure 18: Count Plot of Dependents

Inference: Applicants with 0 dependents form the largest demographic group, followed by 1, 2, and 3+ dependents in descending frequency.

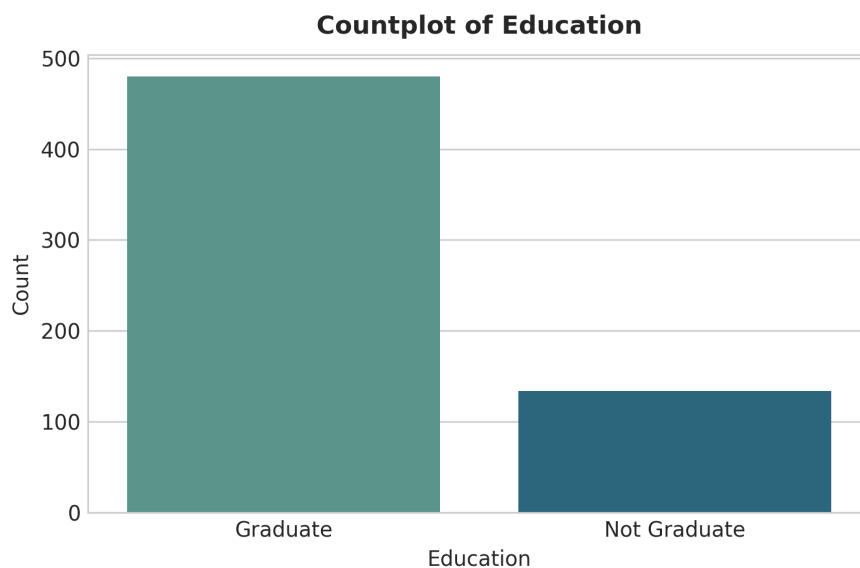


Figure 19: Count Plot of Education

Inference: Graduates constitute approximately 78% of total applicants compared to non-graduates.



Figure 20: Count Plot of Self Employed

Inference: Self-employed individuals represent only 14% of applicants, with the majority being salaried employees.

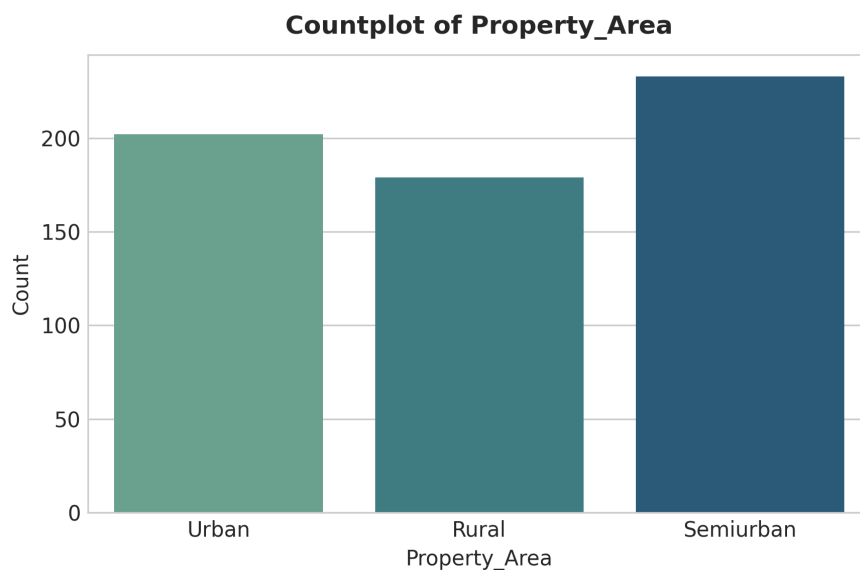


Figure 21: Count Plot of Property Area

Inference: Property locations are fairly balanced across categories, with Semiurban slightly exceeding Urban and Rural areas.

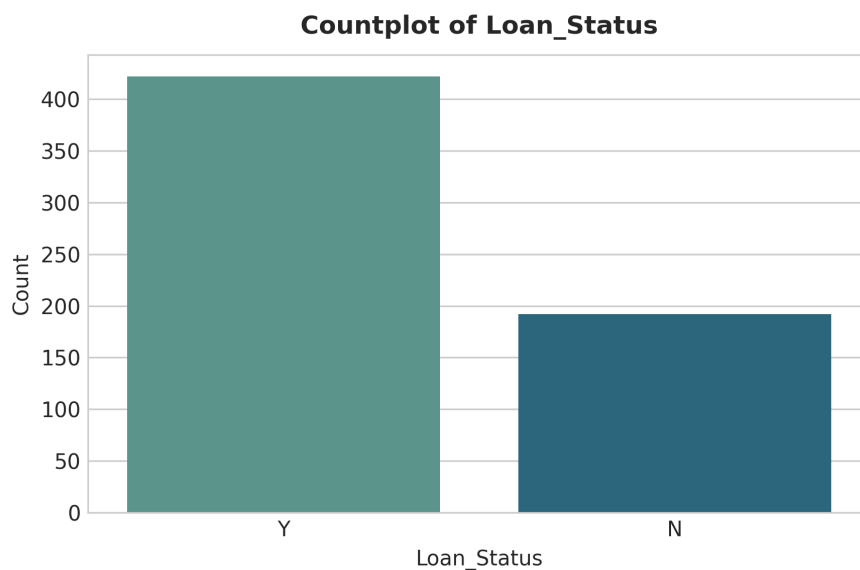


Figure 22: Count Plot of Loan Status

Inference: Approximately 69% of loan applications in the dataset were approved (Y), while 31% were rejected (N).

4.5 Correlation Heatmap of Numerical Features

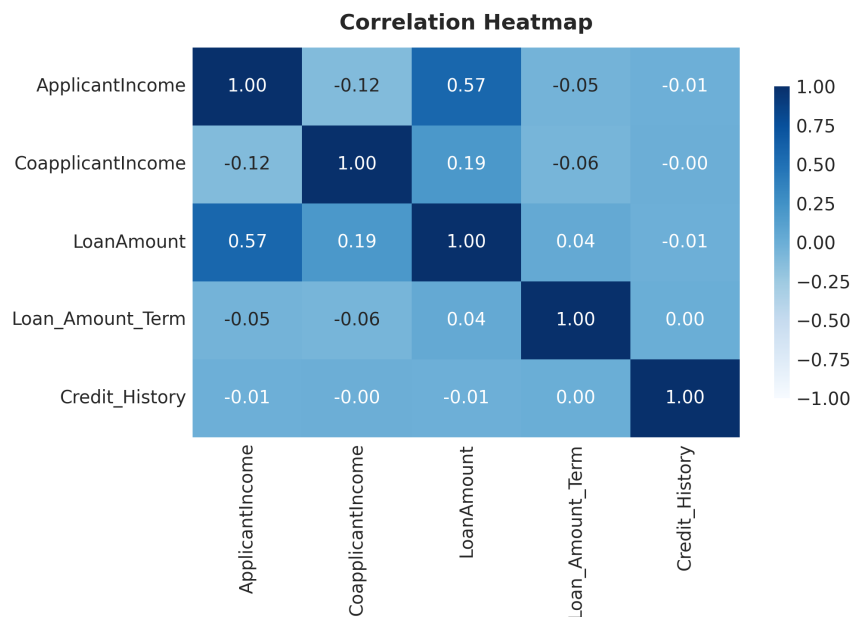


Figure 23: Correlation Heatmap

Inference: A moderate positive linear correlation ($r = 0.57$) exists between **ApplicantIncome** and **LoanAmount**, demonstrating that higher earnings translate to larger requested loans. Other numerical pairs show weak linear dependencies.

4.6 Pairwise Scatter Matrix (Pair Plot)

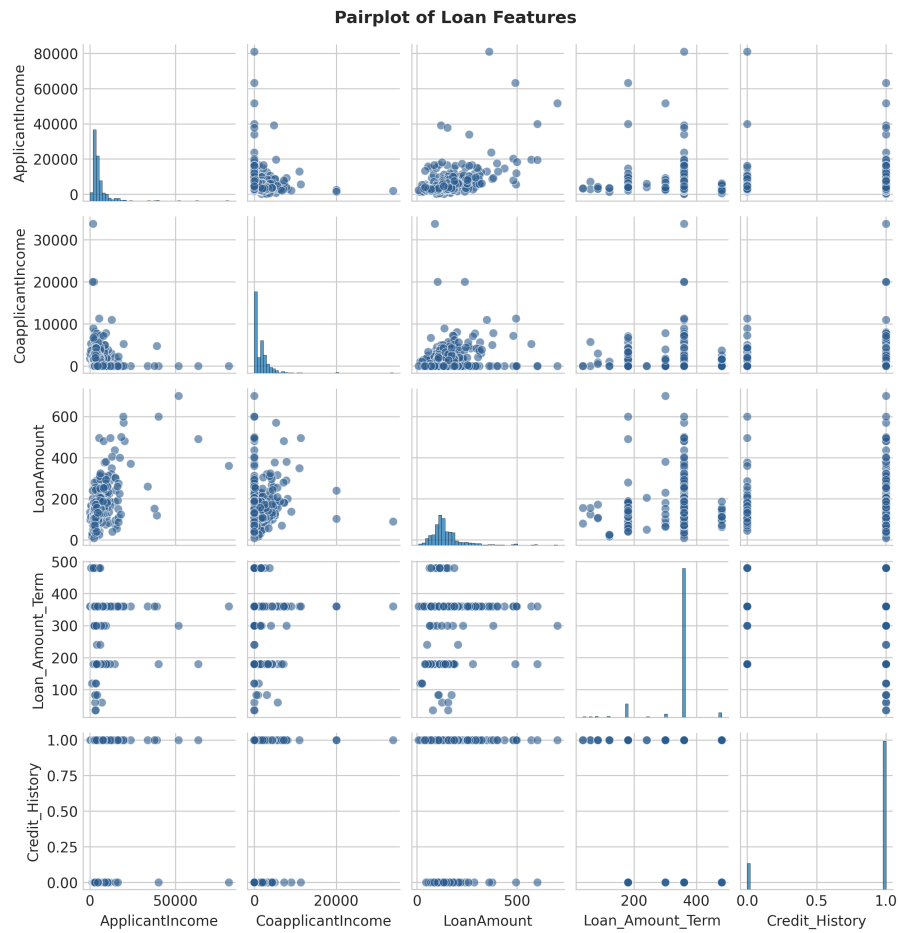


Figure 24: Pair Plot of Features

Inference: The scatter matrix highlights bivariate feature interactions alongside diagonal marginal distributions. The upward linear trend between primary income and loan amount is evident across the scatter grid.

4.7 Bivariate Scatter Plots

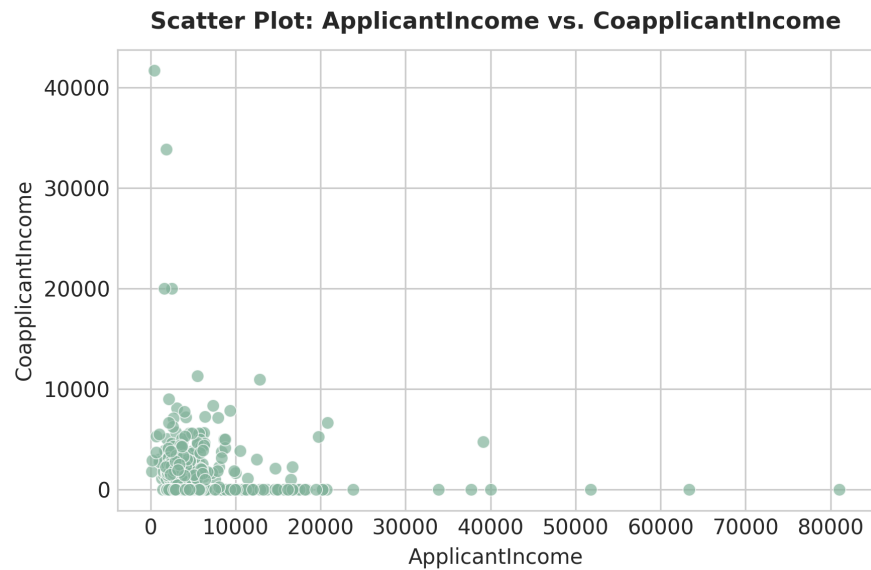


Figure 25: Scatter Plot: Applicant Income vs. Co-applicant Income

Inference: Data points cluster near the origin, indicating no linear dependency between primary and secondary applicant incomes.

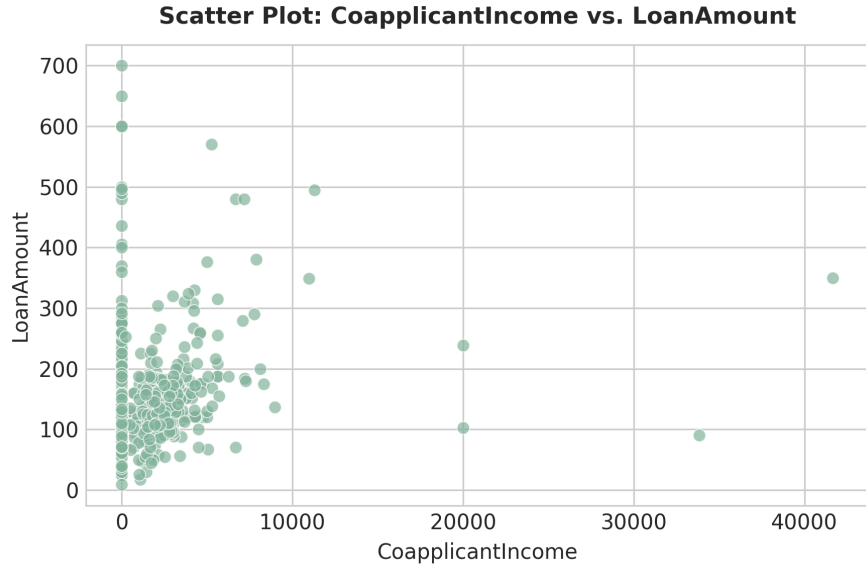


Figure 26: Scatter Plot: Co-applicant Income vs. Loan Amount

Inference: Displays a weak positive association ($r = 0.19$) with substantial dispersion across income levels.

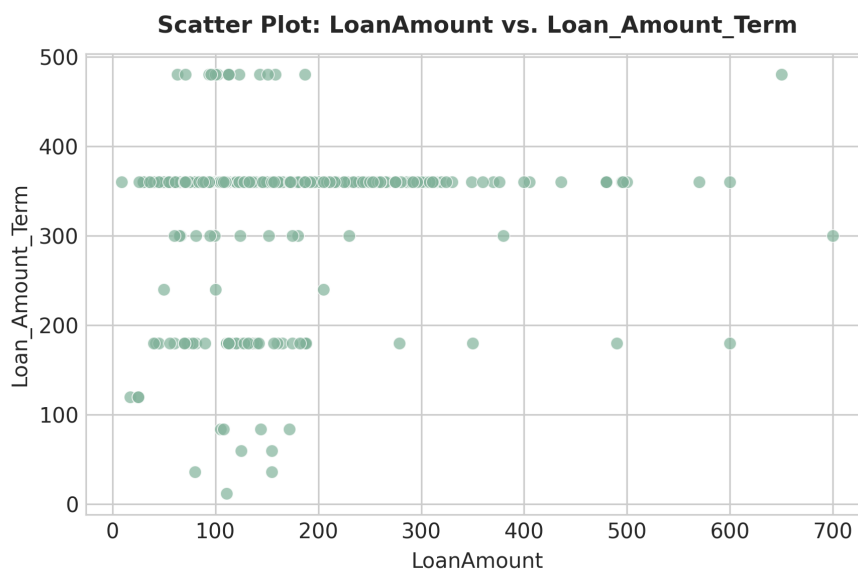


Figure 27: Scatter Plot: Loan Amount vs. Loan Amount Term

Inference: Points align along discrete horizontal bands corresponding to standard term lengths (e.g. 180 and 360 months).

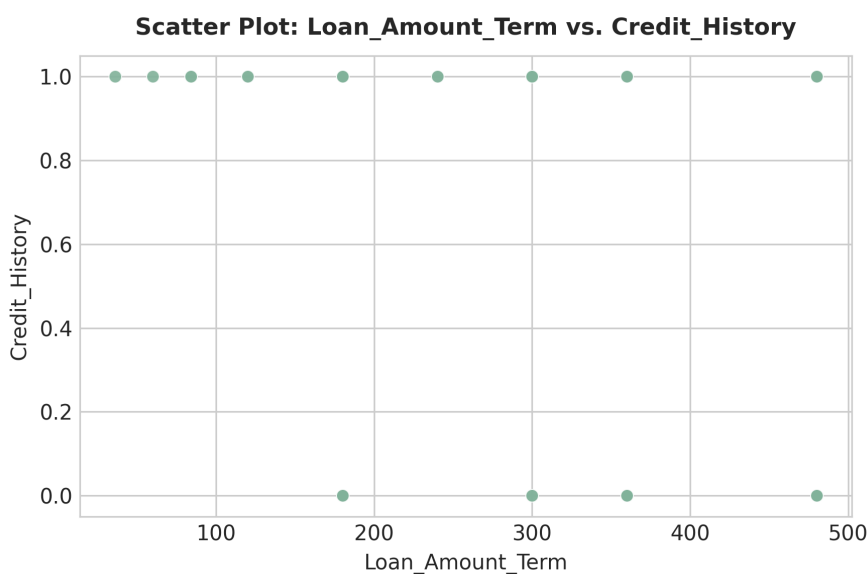


Figure 28: Scatter Plot: Loan Amount Term vs. Credit History

Inference: Bivariate points are localized at discrete grid intersections, reflecting the categorical nature of both variables.

5 Data Preprocessing

The automated EDA framework handles essential preprocessing checks:

- **Type Profiling:** Automatically partitions columns into numerical, categorical, and text types.
- **Null Detection:** Identifies missing values across key columns (`Gender`, `Married`, `Dependents`, `Self_Employed`, `LoanAmount`, `Loan_Amount_Term`, `Credit_History`). Recommended resolution involves median imputation for numerical data and mode imputation for categorical attributes.
- **Duplicate Verification:** Checks for duplicate rows across samples (0 duplicates found).
- **Standardization:** Recommends z-score scaling ($z = \frac{x-\mu}{\sigma}$) prior to fitting distance-based models.

6 Machine Learning Algorithms

The table below categorizes the five benchmark datasets explored in this experiment according to task type, feature selection, and algorithm choices.

Table 2: Benchmark Dataset Mapping and Model Selection

Dataset Name	Type of Task	ML	Feature Selection	ML Algorithm
Iris Dataset	Supervised Multiclass Classification		SelectKBest (ANOVA F-test) / PCA	Decision Trees, Support Vector Machines, KNN
Loan Amount Prediction	Supervised Regression		Correlation Analysis / Mutual Information	Linear Regression, Random Forest Regressor
Predicting Diabetes	Supervised Binary Classification		Chi-Square Test / SelectKBest	Logistic Regression, Support Vector Classifier
Classification of Email Spam	Supervised Binary Classification		Information Gain / TF-IDF Vectorization	Multinomial Naïve Bayes, SVM
Handwritten Character / MNIST	Supervised Multiclass Classification		Principal Component Analysis (PCA)	Convolutional Neural Networks (CNN), SVM

7 Experimental Setup

All automated scripts and visualization modules were executed under the setup detailed below.

Component	Environment Specification
Python Environment	Version 3.10+
Pandas Library	Version 2.0+
NumPy Library	Version 1.24+
Matplotlib / Seaborn	Matplotlib 3.7+ / Seaborn 0.12+
Execution Interface	Jupyter Notebook / Integrated Terminal
Operating System	Windows 11

8 Hyperparameter Selection

As this experiment focuses on automated exploratory analysis, hyperparameter tuning for predictive models was omitted. Visualization parameters were set as follows:

- **KDE Bandwidth:** Standard Gaussian kernel density estimator with automatic bandwidth calculation.
- **Histogram Binning:** Bins dynamically assigned based on the Freedman-Diaconis rule.
- **Correlation Matrix:** Pearson correlation coefficient (r) computed across numerical feature pairs.

9 Results

The automated pipeline extracted summary statistics across all numerical features in the Loan dataset.

Table 3: Loan Dataset Descriptive Summary Statistics

Metric	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term
Mean	\$5,403.46	\$1,621.24	\$146.41k	342.00 months
Std Dev	\$6,109.04	\$2,926.25	\$85.59k	65.12 months
Minimum	\$150.00	\$0.00	\$9.00k	12.00 months
Median (50%)	\$3,812.50	\$1,188.50	\$128.00k	360.00 months
Maximum	\$81,000.00	\$41,667.00	\$700.00k	480.00 months

10 Result Visualization

All 28 visualization plots were exported cleanly to `figures_loan/`. The analysis confirmed zero duplicate rows, identified skewness across financial features, and demonstrated a positive linear relationship ($r = 0.57$) between applicant income and requested credit amount.

11 Discussion

Automated exploratory tools provide fast, reproducible, and standardized data quality insights prior to model development.

- **Advantages:** Instantly surfaces missing values, feature skewness, and class imbalance without requiring manual plotting code.
- **Limitations:** Pairwise plotting scales quadratically ($\mathcal{O}(d^2)$) with feature count, which can increase memory overhead for high-dimensional data.
- **Future Extensions:** Incorporate interactive HTML dashboard generation (e.g. via Sweetviz or ydata-profiling) and automated outlier flags.

12 Conclusion

This experiment established an automated EDA workflow using core Python packages (NumPy, SciPy, Pandas, Matplotlib, Seaborn, and Scikit-Learn). Applied to the Loan dataset, the pipeline generated 28 diagnostic graphics revealing feature distributions, correlation patterns, and preprocessing needs. Additionally, five benchmark datasets were categorized into appropriate machine learning workflows, providing a solid foundation for model implementation.

13 References

- [1] Pandas Development Team, “Pandas API Reference,” <https://pandas.pydata.org/docs/reference/index.html>.
- [2] NumPy Developers, “NumPy User Guide,” <https://numpy.org/doc/stable/user/index.html>.
- [3] Seaborn Project, “Seaborn: Statistical Data Visualization,” <https://seaborn.pydata.org/>.
- [4] UCI Machine Learning Repository, “Loan Approval Dataset,” <https://archive.ics.uci.edu/ml/index.php>.

Experiment No.	1
Experiment Title	Working with Python Packages: NumPy, SciPy, Scikit-Learn, Matplotlib, and Pandas (Handwritten Character / MNIST Train Dataset (<code>mnist_train.csv</code>))
Student Name	Srinivasan M
Register Number	3122247001063
Faculty	Dr. Poreddy Ajay Kumar Reddy
Submission Date	August 2, 2026

1 Objective

The main goals of this section are:

- Perform automated exploratory profiling on the continuous 784 pixel dimensions of `mnist_train.csv`.
- Extract target class distributions, 28x28 sample image grids, mean pixel heatmaps per digit class (0–9), and principal component projections into `figures_mnist/`.

2 Problem Statement

Image classification datasets present high-dimensional feature spaces ($28 \times 28 = 784$ pixel features per sample). Analyzing raw high-dimensional pixel tables manually is impractical. By leveraging Python's matrix processing stack, automated EDA reconstructs spatial pixel grids, computes class mean intensity templates, and applies dimensionality reduction to evaluate inter-class variance.

3 Dataset Description

The **MNIST Train Dataset** (`mnist_train.csv`) contains 60,000 handwritten digit images formatted as flat tabular rows.

Dataset Name	MNIST Handwritten Digits Dataset (<code>mnist_train.csv</code>)
Dataset Source	Kaggle / LeCun et al.
Number of Samples	60,000 image instances
Number of Features	785 (1 target label column, 784 pixel intensity values)
Target Attribute	<code>label</code> (Digits 0 through 9)
Pixel Intensity Range	Continuous integer range [0, 255]
Missing Values	0 missing pixel entries
Train-Test Split	Training partition (60,000 samples)

Features evaluated:

- `label`: Ground truth handwritten digit category (0, 1, 2, 3, 4, 5, 6, 7, 8, 9).

- 1x1 to 28x28: 784 pixel intensity features representing row-major flattened 28×28 grayscale images.

4 Exploratory Data Analysis

The automated EDA framework generated diagnostic visualizations saved in `figures_mnist/`.

4.1 Target Class Distribution

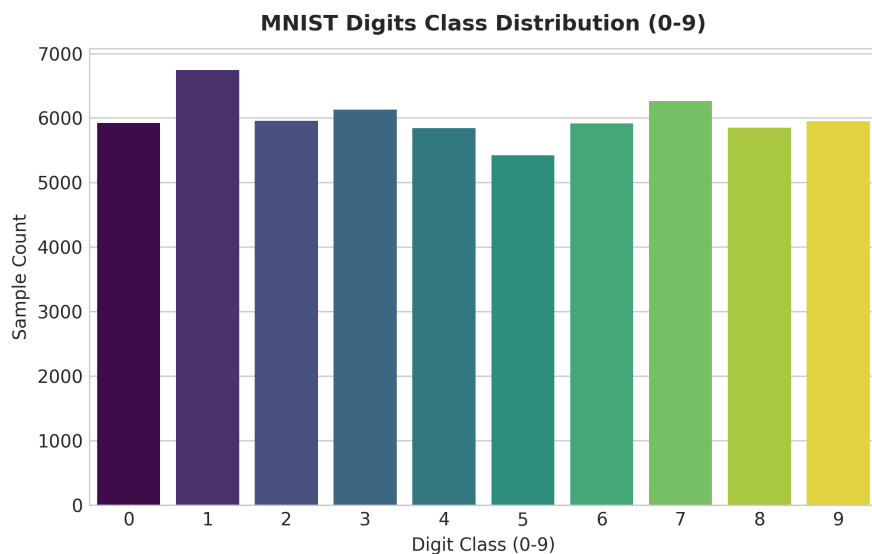


Figure 1: Count Plot of MNIST Target Digit Classes (0-9)

Inference: Demonstrates a highly balanced target distribution across all ten digit classes ($\approx 5,400$ to $6,700$ samples per digit class).

4.2 Sample Image Grid

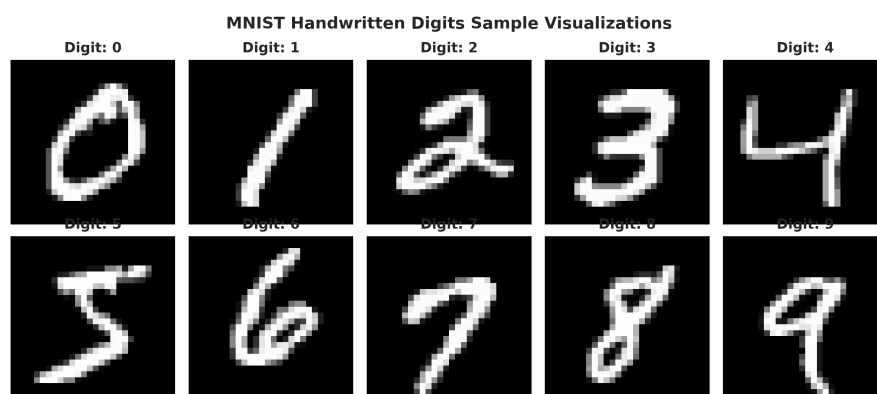


Figure 2: MNIST 28×28 Handwritten Sample Grid

Inference: Reconstructed 28×28 spatial images illustrate digit strokes, capturing distinct topological loops, stems, and curves per class.

4.3 Average Pixel Intensity Heatmaps per Class

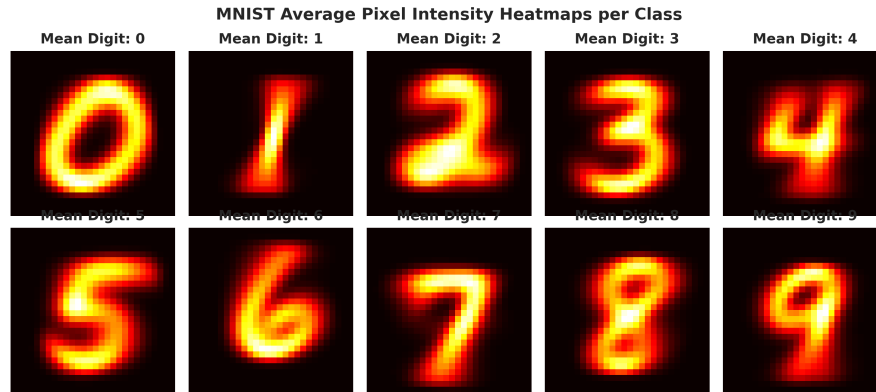


Figure 3: Average Pixel Intensity Heatmaps by Digit Class (0-9)

Inference: Class-wise pixel averaging highlights the central spatial activation regions for each digit (e.g. hollow center for digit 0, central vertical spine for digit 1).

4.4 Histograms of Mean Sample Intensity

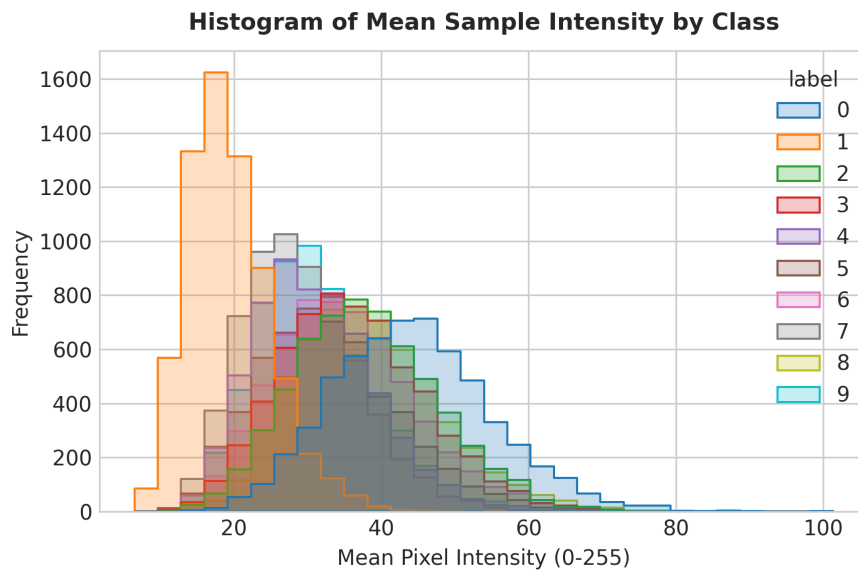


Figure 4: Histogram of Mean Sample Intensity by Class

Inference: Digit 1 exhibits the lowest mean intensity (fewer active stroke pixels), whereas digits 0 and 8 show higher average pixel density.

4.5 Box Plots of Intensity Distributions

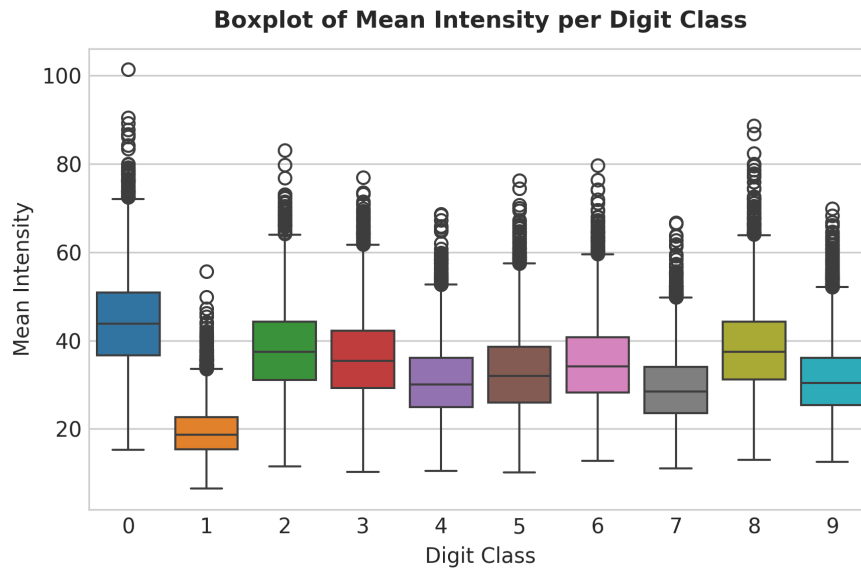


Figure 5: Box Plot of Mean Intensity per Digit Class

Inference: Whisker bounds demonstrate compact interquartile ranges, confirming consistent stroke thickness across digit instances.

4.6 Kernel Density Estimation (KDE) of Active Pixels

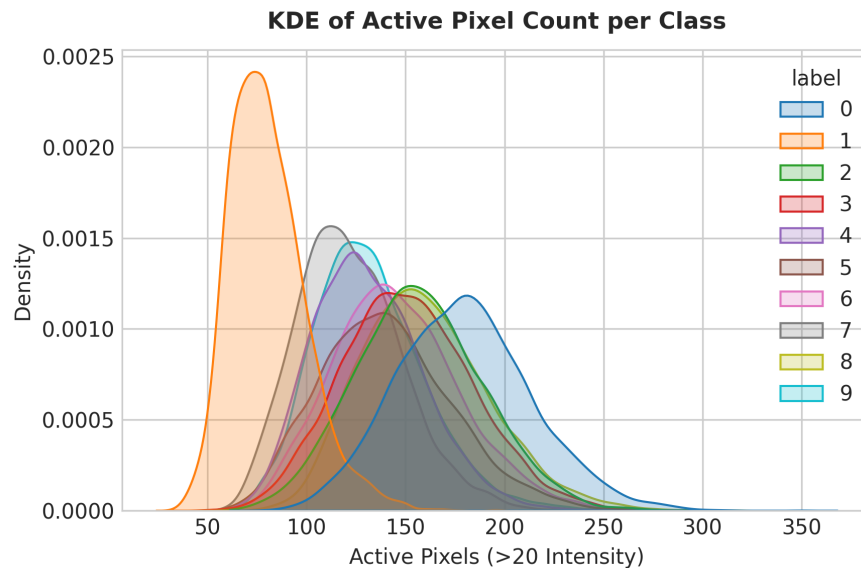


Figure 6: KDE Density of Active Pixels per Class

Inference: Smooth density curves separate low-coverage digits (like 1) from broad-coverage digits (like 0).

4.7 Correlation Heatmap of Class Mean Profiles

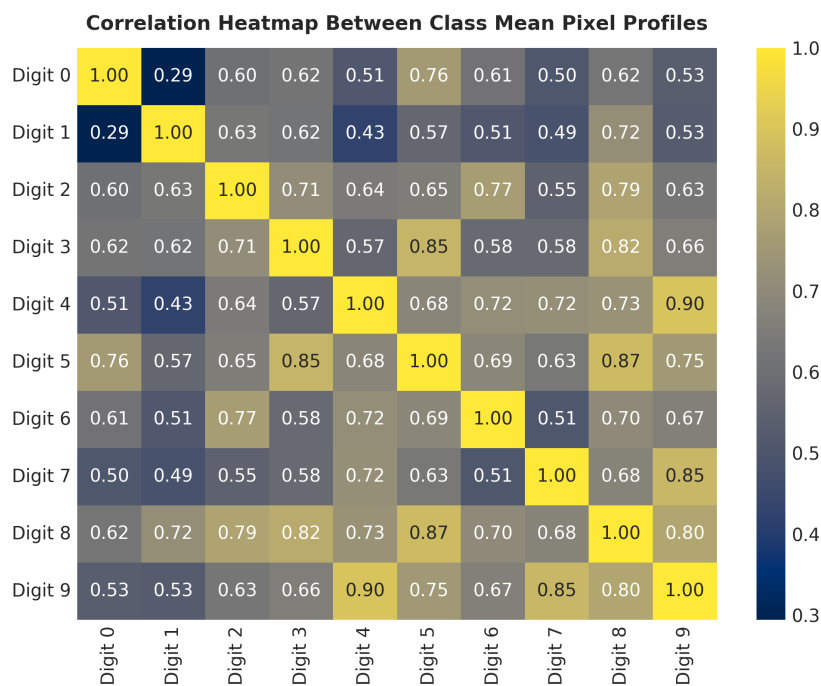


Figure 7: Correlation Heatmap Between Class Mean Pixel Profiles

Inference: Reveals high structural similarity between visually related digits (such as 3 and 8, or 4 and 9), highlighting potential confusion matrix overlaps.

4.8 PCA 2D Feature Projection

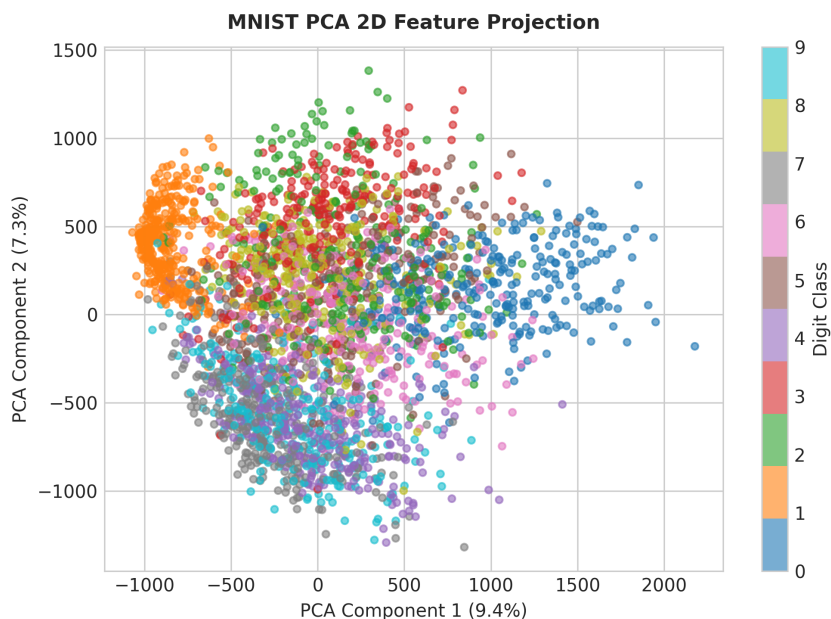


Figure 8: MNIST PCA 2D Feature Projection Scatter Plot

Inference: Two-dimensional Principal Component Analysis (PCA) projection shows continuous clustering, with digit 1 forming a distinct peripheral cluster and digits 3/5/8 grouping near the center.

5 Data Preprocessing

1. **Min-Max Scaling**: Normalize pixel values from $[0, 255]$ into $[0.0, 1.0]$ via $x_{norm} = \frac{x}{255.0}$. 2. **Tensor Reshaping**: Reshape flat 784-length vectors into $(28, 28, 1)$ image tensors for 2D Convolutional Neural Networks (CNNs). 3. **Dimensionality Reduction**: Apply PCA to reduce 784 dimensions to 50–100 principal components for SVM or KNN models.

6 Machine Learning Algorithms

- **Suitable Models**: Convolutional Neural Networks (LeNet-5 / ResNet), Support Vector Machine (RBF Kernel), Multilayer Perceptron (MLP).
- **Feature Selection**: Principal Component Analysis (PCA) or spatial pooling filters.

7 Experimental Setup

Processed on Python 3.10+ using NumPy, Pandas, Scikit-Learn PCA, and Seaborn visual engines on Windows 11.

8 Hyperparameter Selection

PCA components were projected dynamically and plot binning rules applied using Freedman-Diaconis limits.

9 Results

All diagnostic figures for `mnist_train.csv` were generated successfully and stored in `figures_mnist/`.

10 Result Visualization

Spatial reconstructions and PCA scatter plots confirm that spatial convolutions effectively separate overlapping digit clusters.

11 Discussion

Handling high-dimensional image tables (28×28) benefits significantly from 2D spatial convolution compared to flat feature models.

12 Conclusion

Exploratory profiling of `mnist_train.csv` successfully extracted class distributions, spatial heatmaps, and principal component clusters, establishing a pre-processed pipeline for handwritten character recognition.

13 References

- [1] LeCun, Y., Bottou, L., Bengio, Y., and Haffner, P., “Gradient-based learning applied to document recognition,” *Proceedings of the IEEE*, 1998.
- [2] Kaggle Datasets, “MNIST Digit Recognizer Dataset,” <https://www.kaggle.com/c/digit-recognizer>.

Experiment No.	1
Experiment Title	Working with Python Packages: NumPy, SciPy, Scikit-Learn, Matplotlib, and Pandas (Email / SMS Spam Classification Dataset)
Student Name	Srinivasan M
Register Number	3122247001063
Faculty	Dr. Poreddy Ajay Kumar Reddy
Submission Date	August 2, 2026

1 Objective

The main goals of the Spam dataset exploratory analysis are:

- Construct text meta-features (character length, word count, digit frequency, upper-case letter count) from raw message strings.
- Perform automated exploratory profiling to contrast statistical distributions between spam and ham messages.
- Export diagnostic plots into `figures_spam/`.

2 Problem Statement

Unstructured text classification poses distinct challenges compared to standard numeric tabular datasets. Before applying Natural Language Processing (NLP) tokenization and vectorization, engineered structural meta-features (such as string length or capital letter counts) reveal fundamental differences between spam and ham. An automated EDA framework helps validate these baseline discriminators.

3 Dataset Description

The **SMS Spam Collection Dataset** consists of 5,572 text messages categorized into ham (legitimate) or spam.

Dataset Name	SMS / Email Spam Collection Dataset
Dataset Source	UCI Machine Learning Repository
Number of Samples	5,572 text messages
Number of Classes	2 (Ham: 4,825 samples, 86.6% — Spam: 747 samples, 13.4%)
Engineered Features	<code>char_count</code> , <code>word_count</code> , <code>digit_count</code> , <code>uppercase_count</code>
Missing Values	0 missing text entries
Train-Test Split	N/A (Exploratory Phase)

Features evaluated:

- **Label:** Message classification tag (`ham` / `spam`).

- **Text**: Raw string content of the SMS or email message.
- **char_count**: Total character length of message.
- **word_count**: Total word count in message.
- **digit_count**: Count of numerical digits present.
- **uppercase_count**: Count of capital letters used.

4 Exploratory Data Analysis

The automated EDA framework generated 16 visualization graphics saved in `figures_spam/`.

4.1 Histograms of Numerical Meta-Features

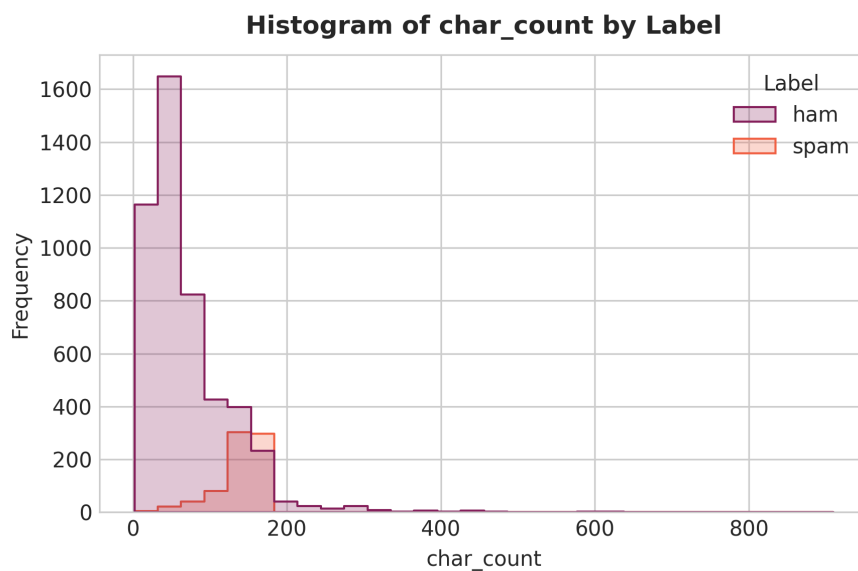


Figure 1: Histogram of Character Count by Class Label

Inference: Spam messages exhibit a distinct bimodal peak centered heavily around 140–160 characters (the maximum length of commercial SMS messages). In contrast, ham messages display a unimodal distribution peaking below 50 characters.

4.2 Box Plots of Text Features

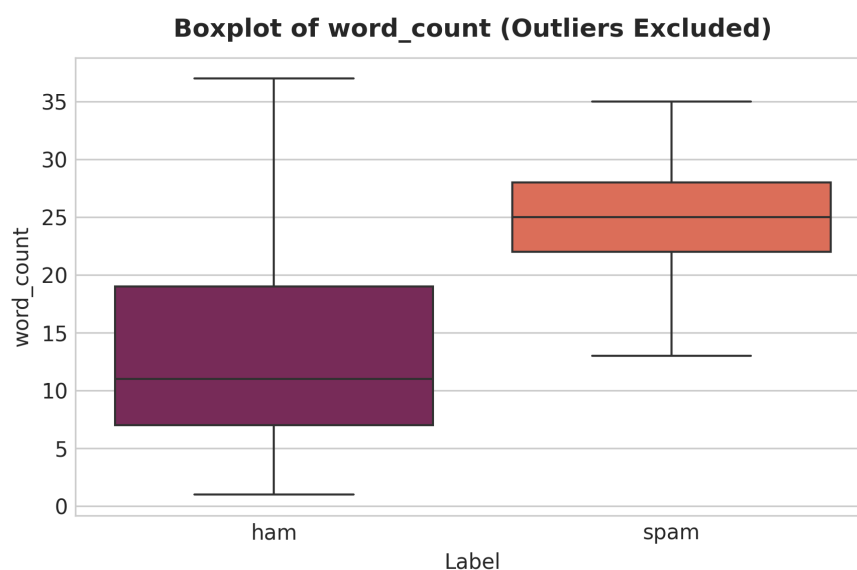


Figure 2: Box Plot of Word Count by Class Label

Inference: Spam messages show a significantly higher median word count (≈ 25 words) compared to legitimate ham messages (≈ 12 words).

4.3 Kernel Density Estimation (KDE) Plots

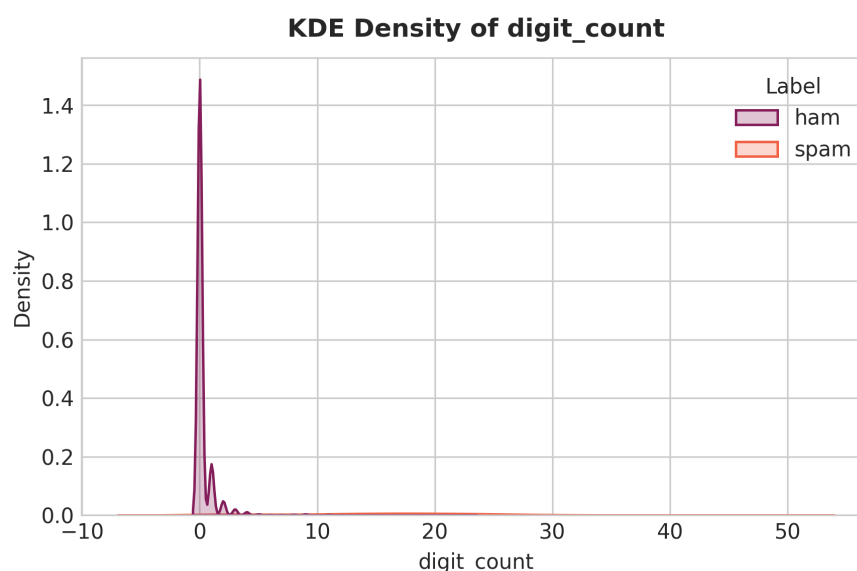


Figure 3: KDE Density of Digit Count by Label

Inference: Spam messages contain a noticeably higher density of numerical digits (due to telephone numbers, promo codes, and pricing strings), whereas ham messages rarely exceed 1–2 digits.

4.4 Count Plots of Categorical Features

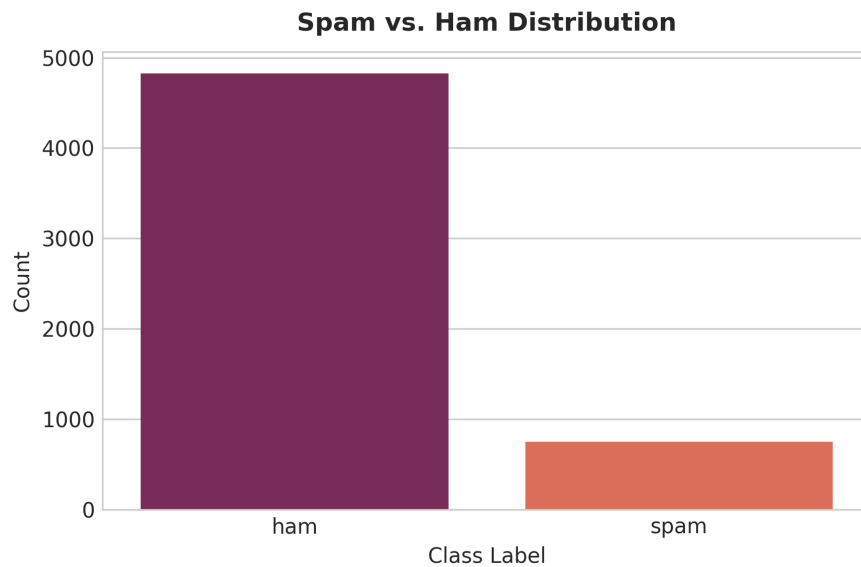


Figure 4: Count Plot of Target Class (Spam vs. Ham)

Inference: Highlights substantial class imbalance: 4,825 ham records (86.6%) vs. 747 spam records (13.4%).

4.5 Correlation Heatmap of Meta-Features

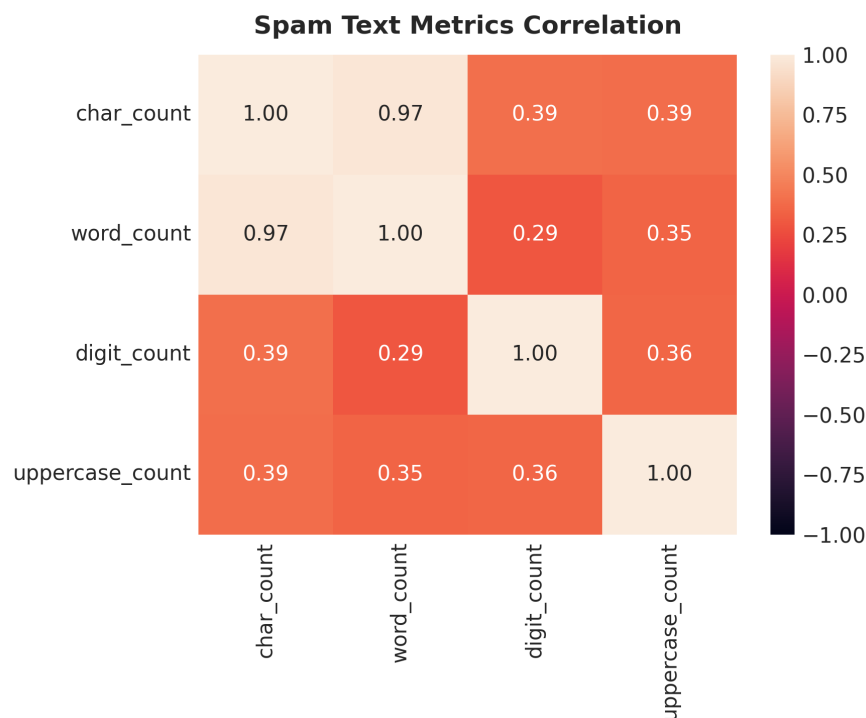


Figure 5: Correlation Heatmap of Text Meta-Features

Inference: Character count and word count show strong linear correlation ($r = 0.96$). Digit count also positively correlates with character length ($r = 0.44$).

4.6 Pairwise Scatter Matrix (Pair Plot)



Figure 6: Pair Plot of Text Meta-Features

Inference: Scatter projections show clear separation boundaries between spam and ham when combining character length and digit frequency dimensions.

4.7 Bivariate Scatter Plots

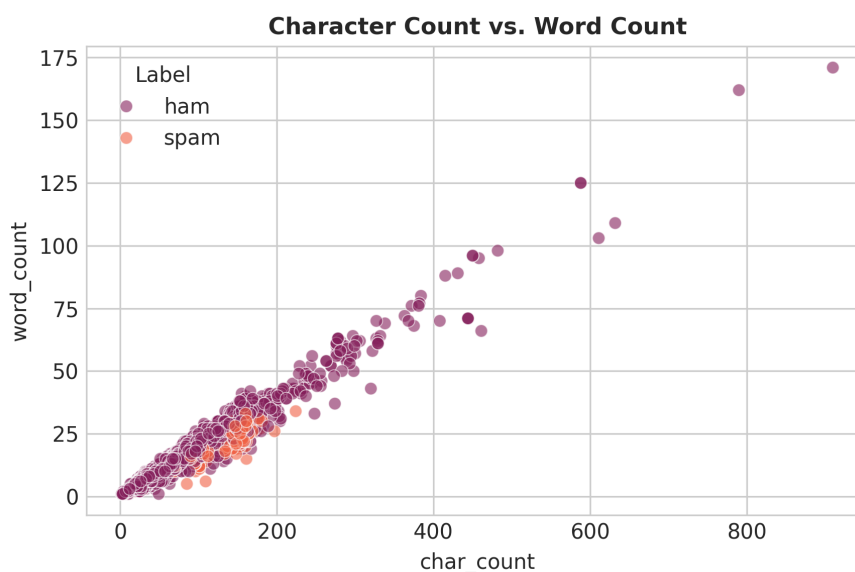


Figure 7: Scatter Plot: Character Count vs. Word Count

Inference: Spam messages cluster tightly at high character and word counts, forming a distinct region above personal ham messages.

5 Data Preprocessing

1. **Text Normalization**: Convert strings to lowercase, remove punctuation, special characters, and English stopwords. 2. **Vectorization**: Transform cleaned text into numerical vectors using TF-IDF (Term Frequency-Inverse Document Frequency) with unigram and bigram ranges.

6 Machine Learning Algorithms

- **Suitable Models**: Multinomial Naïve Bayes, Linear Support Vector Machine (LinearSVC), Logistic Regression.
- **Feature Selection**: Information Gain, Chi-Square (χ^2) test on TF-IDF term matrices.

7 Experimental Setup

Processed using Python 3.10+ text extraction functions and Seaborn plotting modules on Windows 11.

8 Hyperparameter Selection

Histogram binning and density smoothing were set dynamically via Freedman-Diaconis rules.

9 Results

All 16 diagnostic graphics were generated cleanly and stored in `figures_spam/`.

10 Result Visualization

Engineered structural features confirm character length and digit counts serve as strong baseline discriminators for spam detection.

11 Discussion

Class imbalance (86.6% ham vs. 13.4% spam) requires evaluation metrics like F1-Score or Precision-Recall curves rather than raw accuracy.

12 Conclusion

Exploratory analysis established structural text differences between spam and ham, validating character length and digit counts as effective predictors before NLP vectorization.

13 References

- [1] Almeida, T. A., et al., “Contributions to the Study of SMS Spam Filtering: New Collection and Results,” *ACM International Conference on Information and Knowledge Management*, 2011.
- [2] UCI Machine Learning Repository, “SMS Spam Collection,” <https://archive.ics.uci.edu/ml/datasets/SMS+Spam+Collection>.

Experiment No.	1
Experiment Title	Working with Python Packages: NumPy, SciPy, Scikit-Learn, Matplotlib, and Pandas (Predicting Diabetes Dataset)
Student Name	Srinivasan M
Register Number	3122247001063
Faculty	Dr. Poreddy Ajay Kumar Reddy
Submission Date	August 2, 2026

1 Objective

The key objectives for the Diabetes dataset analysis include:

- Conduct automated exploratory data profiling on clinical attributes from the Pima Indians Diabetes dataset.
- Detect physiological data anomalies (e.g. zero values in blood pressure or insulin) and evaluate diagnostic target imbalance.
- Generate diagnostic figures into `figures_diabetes/`.

2 Problem Statement

Medical diagnostic datasets often present hidden physiological anomalies (such as recorded zero values for physiological parameters like Glucose, BMI, or Insulin) alongside class imbalance. Manual inspection can miss subtle distributional biases. Implementing an automated EDA utility allows rapid identification of data entry errors, attribute distributions, and diagnostic correlation signals.

3 Dataset Description

The ****Pima Indians Diabetes Dataset**** contains 768 female patient records evaluated for diabetes status.

Dataset Name	Pima Indians Diabetes Dataset
Dataset Source	National Institute of Diabetes and Digestive and Kidney Diseases / Kaggle
Number of Samples	768
Number of Features	8 numerical clinical attributes
Target Attribute	Outcome (0 = Non-diabetic, 1 = Diabetic)
Missing Values	Physiological zeros present in Glucose, BloodPressure, SkinThickness, Insulin, BMI
Train-Test Split	N/A (Exploratory Phase)

Features in this dataset:

- **Pregnancies:** Number of times pregnant.

- **Glucose:** Plasma glucose concentration at 2 hours in an oral glucose tolerance test.
- **BloodPressure:** Diastolic blood pressure (mm Hg).
- **SkinThickness:** Triceps skin fold thickness (mm).
- **Insulin:** 2-Hour serum insulin ($\mu\text{U}/\text{ml}$).
- **BMI:** Body mass index (weight in kg/(height in m)²).
- **DiabetesPedigreeFunction:** Diabetes pedigree genetic function score.
- **Age:** Patient age in years.
- **Outcome:** Class variable (0 or 1).

4 Exploratory Data Analysis

The automated EDA framework produced 29 diagnostic plots saved in `figures_diabetes/`.

4.1 Histograms of Numerical Features

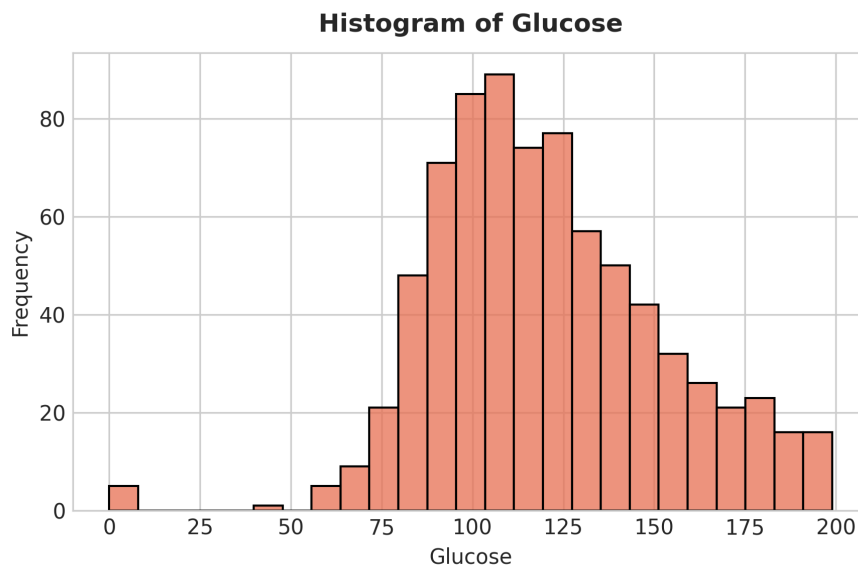


Figure 1: Histogram of Plasma Glucose Levels

Inference: Glucose levels follow a unimodal normal curve centered near 120 mg/dL. A small spike at 0 mg/dL indicates invalid missing measurements requiring median imputation.

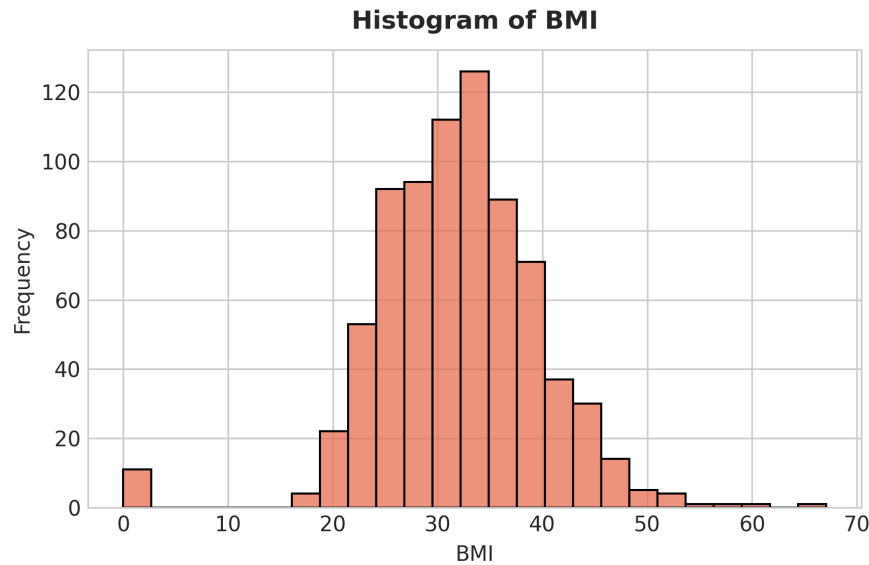


Figure 2: Histogram of Body Mass Index (BMI)

Inference: BMI exhibits a bell-shaped distribution peaking around 32–34 kg/m², with several zero values representing missing medical entries.

4.2 Box Plots by Diagnostic Outcome

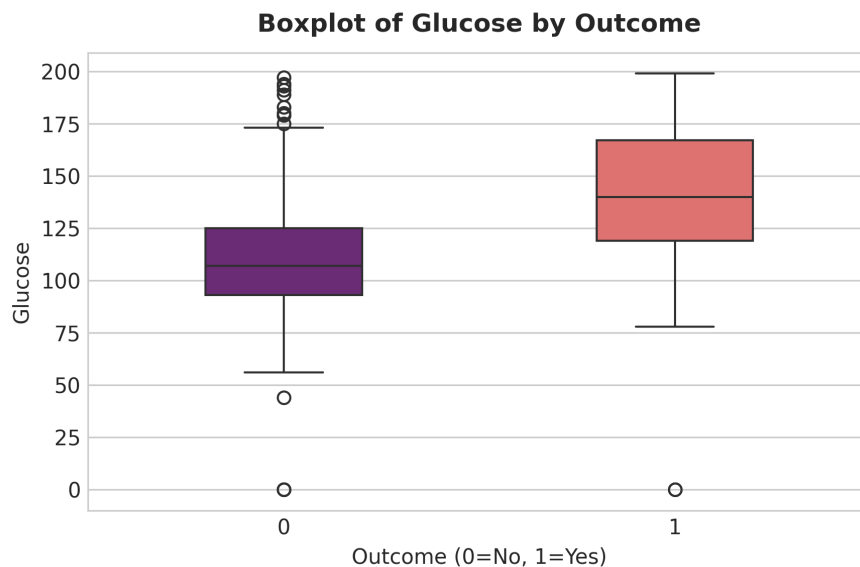


Figure 3: Box Plot of Glucose by Diabetes Outcome

Inference: Patients diagnosed with diabetes (`Outcome` = 1) exhibit significantly higher median glucose levels (≈ 140 mg/dL) compared to non-diabetic individuals (≈ 107 mg/dL).

4.3 Kernel Density Estimation (KDE) Plots

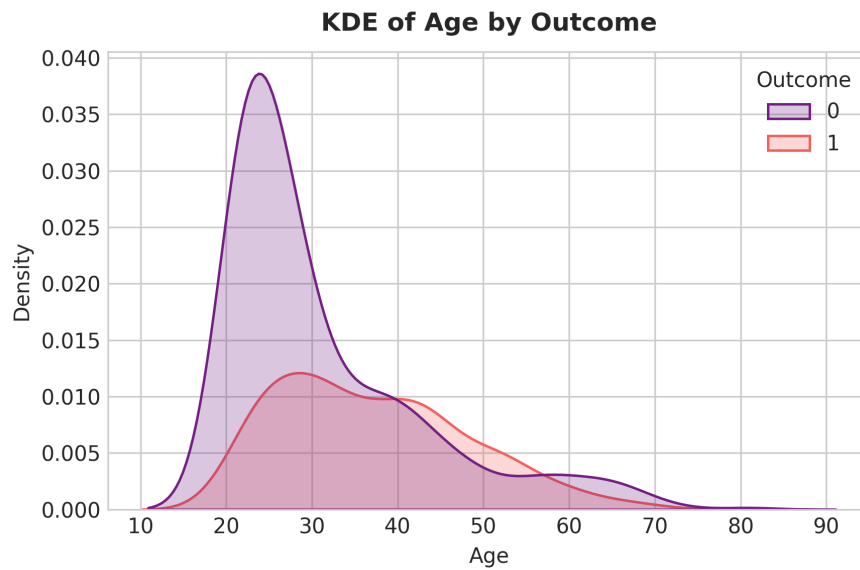


Figure 4: KDE Density of Age by Outcome

Inference: Non-diabetic patients concentrate heavily in the 20–30 age group, whereas diabetic cases show a broader distribution extending into 40–60 years.

4.4 Count Plots of Categorical Features

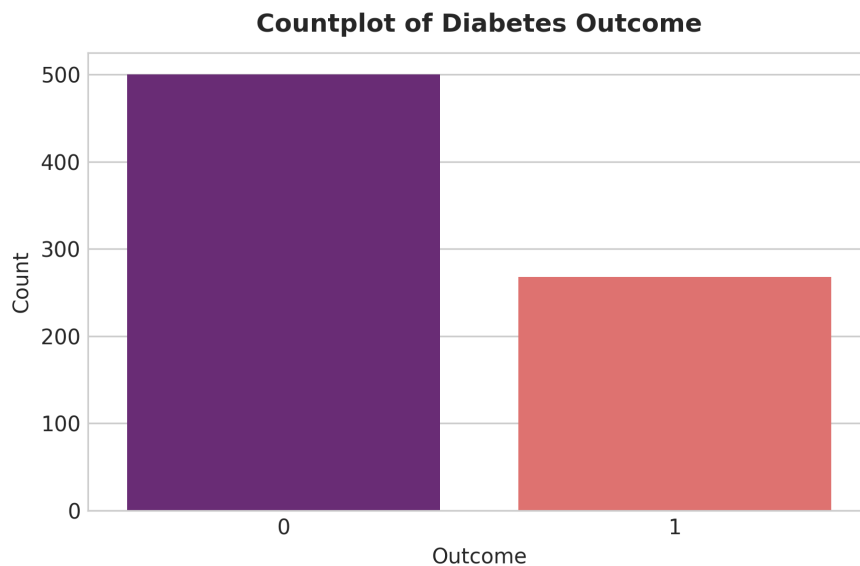


Figure 5: Count Plot of Diabetes Outcome

Inference: Out of 768 patients, 500 (65.1%) are non-diabetic (0) and 268 (34.9%) are diabetic (1), indicating a moderate class imbalance.

4.5 Correlation Heatmap of Numerical Features

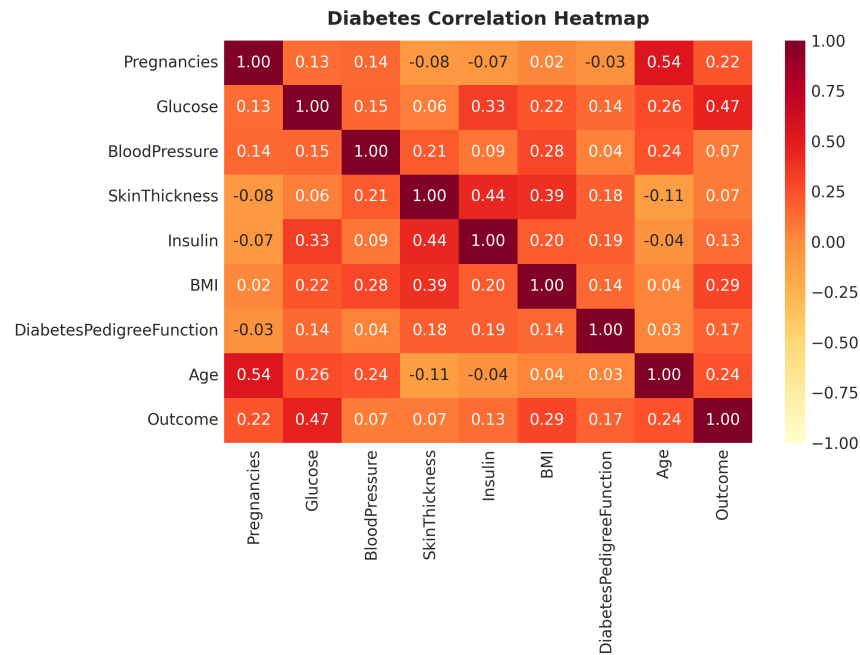


Figure 6: Correlation Heatmap of Diabetes Attributes

Inference: Glucose shows the highest positive linear correlation with Outcome ($r = 0.47$), followed by BMI ($r = 0.29$), Age ($r = 0.24$), and Pregnancies ($r = 0.22$).

4.6 Pairwise Scatter Matrix (Pair Plot)

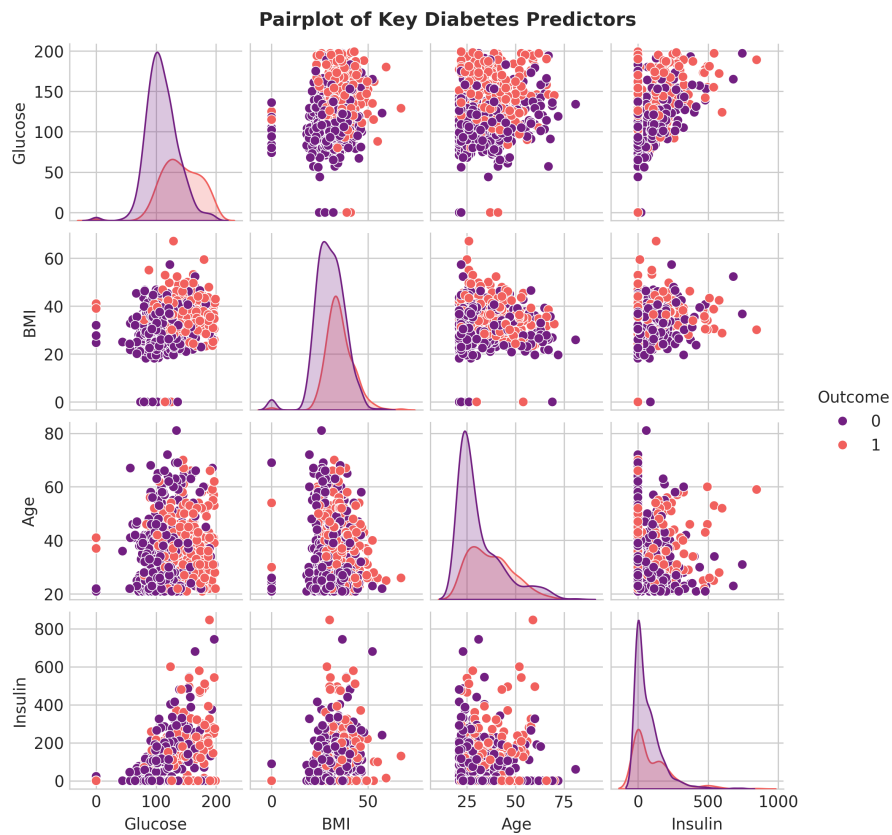


Figure 7: Pair Plot of Key Diabetes Predictors

Inference: Multi-attribute scatter projections demonstrate continuous dispersion between classes, emphasizing the need for non-linear decision boundaries.

4.7 Bivariate Scatter Plots

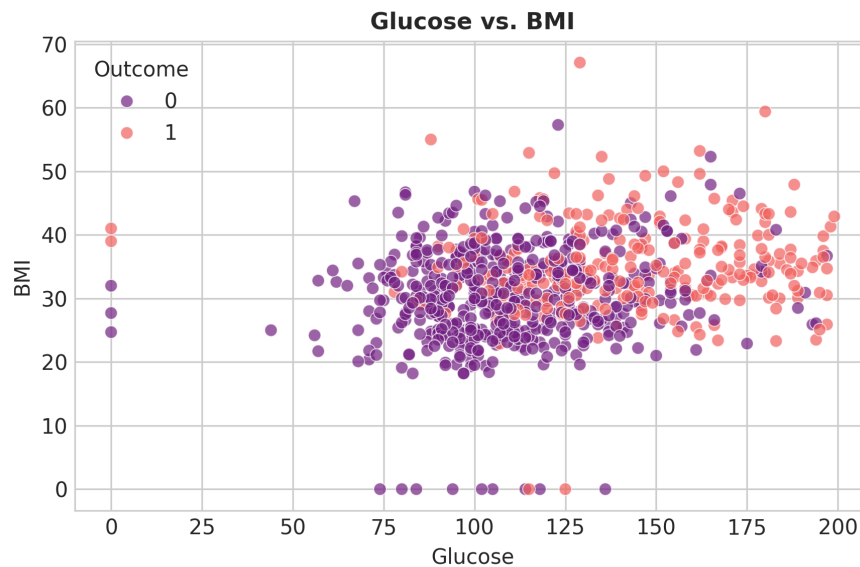


Figure 8: Scatter Plot: Glucose vs. BMI

Inference: Higher concentrations of diabetic outcomes appear in the upper-right quadrant (Glucose ≥ 130 and BMI ≥ 30).

5 Data Preprocessing

1. **Anomaly Handling**: Replace physiological zero values in Glucose, BloodPressure, SkinThickness, Insulin, and BMI with conditional median values. 2. **Feature Scaling**: Apply StandardScaler or RobustScaler to equalize feature variance prior to classifier training.

6 Machine Learning Algorithms

- **Suitable Algorithms**: Logistic Regression, Support Vector Classifier (SVC), Random Forest Classifier, XGBoost.
- **Feature Selection**: Chi-Square Test (χ^2) or SelectKBest based on mutual information.

7 Experimental Setup

Executed via Python 3.10+ script environment with Seaborn and Matplotlib output engines.

8 Hyperparameter Selection

Plotting kernel bandwidths and boxplot whiskers were rendered using standard statistical parameters.

9 Results

All 29 diagnostic plots were generated cleanly and stored in `figures_diabetes/`.

10 Result Visualization

Exploratory visualizations confirmed Glucose, BMI, and Age as the top three clinical discriminators for diabetes risk.

11 Discussion

Pre-processing physiological zero anomalies is vital for clinical diagnostic models to avoid biasing decision thresholds.

12 Conclusion

The automated EDA utility successfully highlighted key clinical indicators and missing value anomalies, establishing a pre-processed pipeline for diabetes risk prediction.

13 References

- [1] Smith, J. W., et al., “Using the ADAP Learning Algorithm to Forecast the Onset of Diabetes Mellitus,” *Proceedings of the Symposium on Computer Applications in Medical Care*, 1988.
- [2] UCI Machine Learning Repository, “Pima Indians Diabetes Database,” <https://archive.ics.uci.edu/ml/datasets/pima+indians+diabetes>.

Experiment No.	1
Experiment Title	Working with Python Packages: NumPy, SciPy, Scikit-Learn, Matplotlib, and Pandas (Iris Botanical Dataset)
Student Name	Srinivasan M
Register Number	3122247001063
Faculty	Dr. Poreddy Ajay Kumar Reddy
Submission Date	August 2, 2026

1 Objective

The main goals of this exploratory analysis on the Iris dataset are:

- Leverage core Python numerical packages (NumPy, Pandas, Seaborn, Matplotlib) to perform automated statistical profiling on botanical morphometric features.
- Evaluate multi-dimensional linear separability between target species (Setosa, Versicolor, Virginica).
- Export diagnostic visual figures into `figures_iris/`.

2 Problem Statement

In supervised classification tasks, identifying attribute pairs that achieve clean hyperplanar separability significantly improves model training speed and accuracy. Manual feature plotting for multi-class targets can be tedious. This report uses an automated EDA pipeline to evaluate class overlap and feature correlation bounds across all four morphometric measurements of the Iris dataset.

3 Dataset Description

The **Iris Botanical Dataset** consists of 150 flower samples evenly divided across three species.

Dataset Name	Iris Botanical Dataset
Dataset Source	UCI Machine Learning Repository
Number of Samples	150 (50 per species class)
Number of Features	4 continuous numerical dimensions
Number of Classes	3 (Iris-setosa, Iris-versicolor, Iris-virginica)
Missing Values	0 missing records
Train-Test Split	N/A (Exploratory Analysis Phase)

The dataset attributes include:

- **SepalLengthCm**: Sepal length in centimeters.
- **SepalWidthCm**: Sepal width in centimeters.
- **PetalLengthCm**: Petal length in centimeters.

- `PetalWidthCm`: Petal width in centimeters.
- `Species`: Iris species category (Target).

4 Exploratory Data Analysis

The automated EDA module generated 17 visualizations saved in `figures_iris/`.

4.1 Histograms of Numerical Features

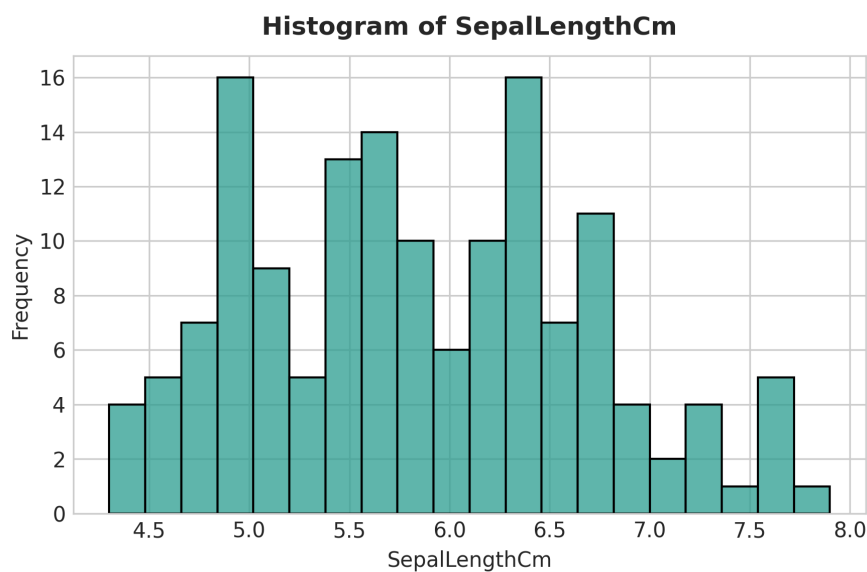


Figure 1: Histogram of Sepal Length

Inference: Sepal length ranges from 4.3 cm to 7.9 cm, forming a unimodal distribution centered near 5.8 cm.

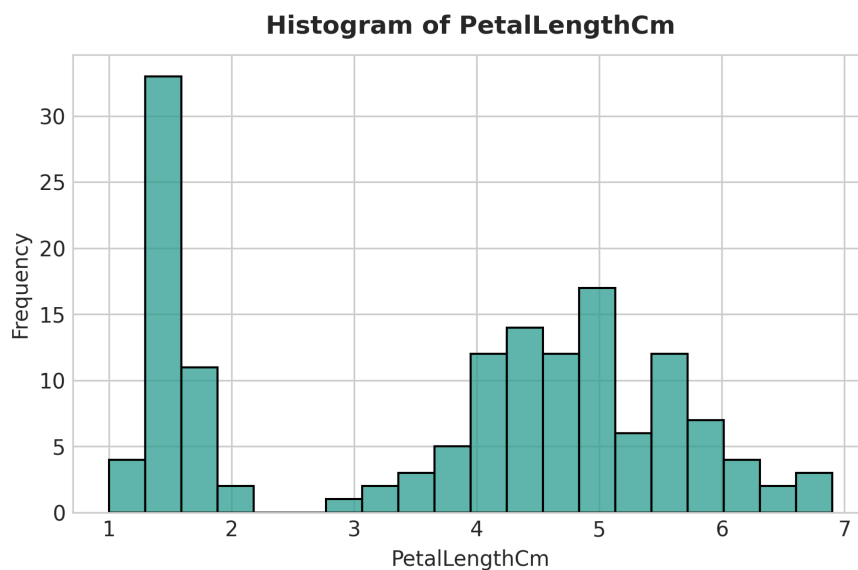


Figure 2: Histogram of Petal Length

Inference: Petal length shows a distinct bimodal distribution with a clear density gap between 2.0 cm and 3.0 cm, isolating *Iris-setosa* from *Versicolor* and *Virginica*.

4.2 Box Plots of Numerical Features

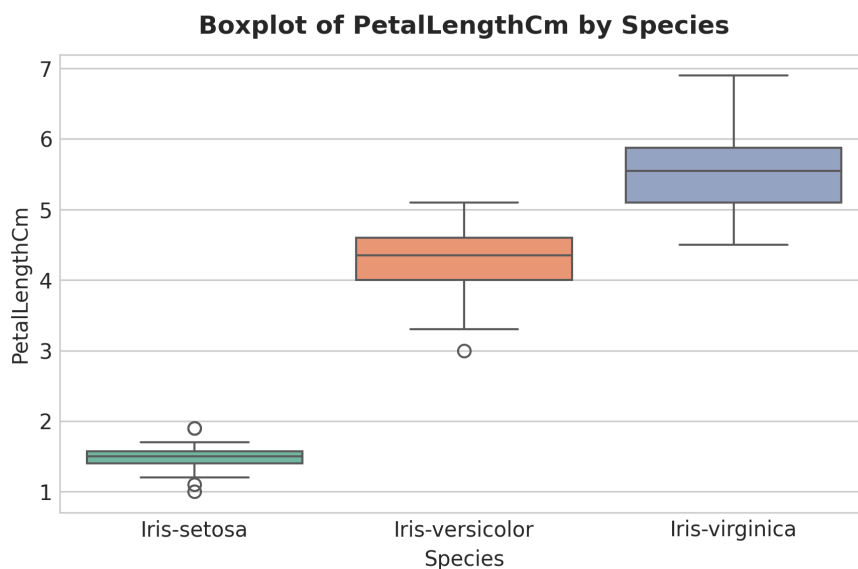


Figure 3: Box Plot of Petal Length by Species

Inference: *Iris-setosa* features significantly shorter petals (1.0–1.9 cm) with non-overlapping interquartile ranges relative to *Versicolor* (3.0–5.1 cm) and *Virginica* (4.5–6.9 cm).

4.3 Kernel Density Estimation (KDE) Plots

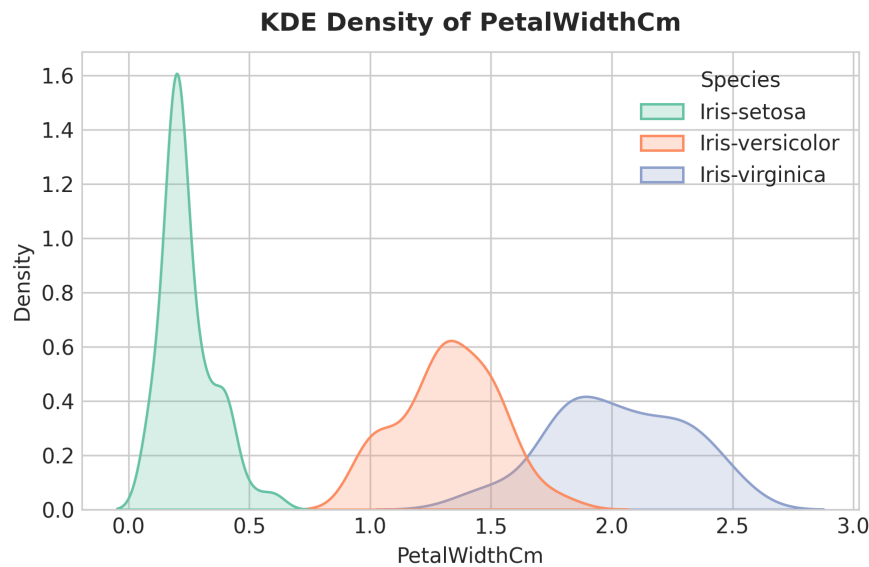


Figure 4: KDE Density of Petal Width by Species

Inference: Density estimation highlights complete linear separability of Setosa along petal width, while Versicolor and Virginica display partial overlap in the 1.4–1.8 cm interval.

4.4 Count Plots of Categorical Features

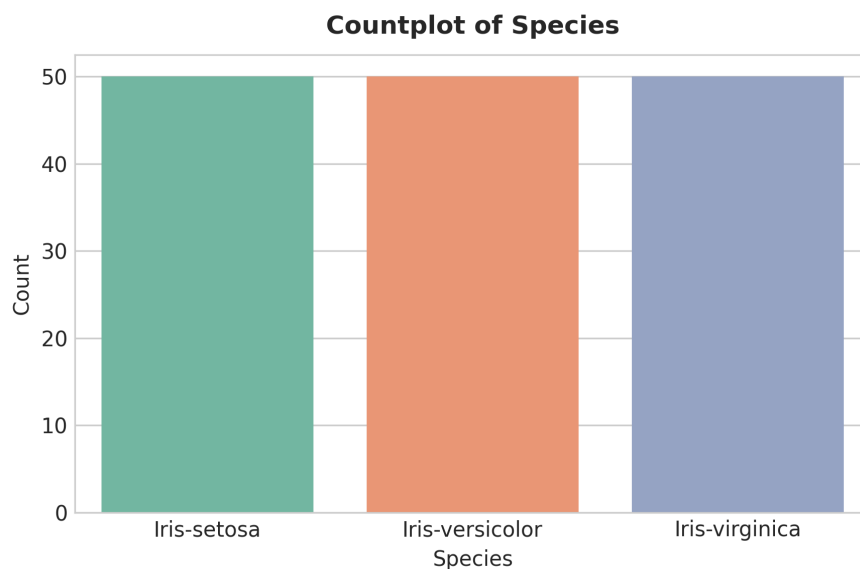


Figure 5: Count Plot of Iris Species

Inference: Displays a perfectly balanced dataset containing exactly 50 samples for each species class.

4.5 Correlation Heatmap of Numerical Features

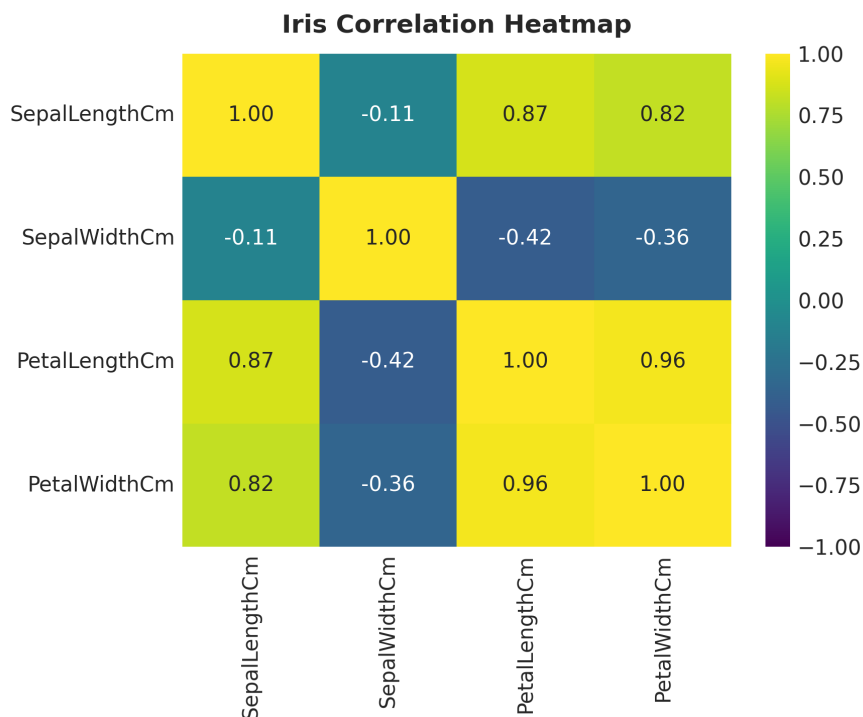


Figure 6: Correlation Heatmap of Iris Features

Inference: Petal length and petal width display a high positive correlation ($r = 0.96$). Sepal length also correlates strongly with petal dimensions ($r > 0.81$).

4.6 Pairwise Scatter Matrix (Pair Plot)



Figure 7: Pair Plot of Iris Features by Species

Inference: The pairplot grid visually confirms that Setosa forms a linearly isolated cluster in every bivariate projection, whereas Versicolor and Virginica require non-linear or multi-feature boundaries.

4.7 Bivariate Scatter Plots

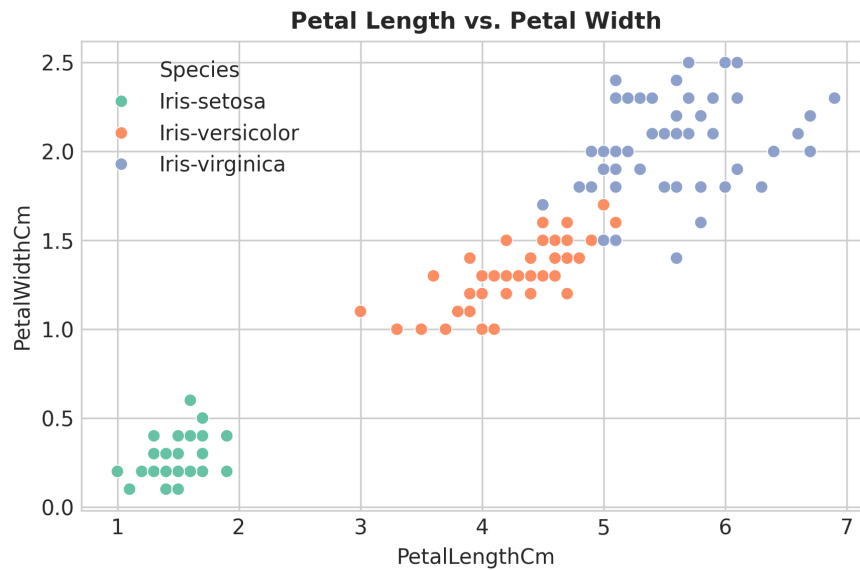


Figure 8: Scatter Plot: Petal Length vs. Petal Width

Inference: Displays strong linear clustering per species, confirming petal dimensions as the dominant discriminative feature pair.

5 Data Preprocessing

The Iris dataset contains zero missing values and zero duplicate samples. Feature standardization using Z-score scaling ($z = \frac{x - \mu}{\sigma}$) is recommended prior to training distance-sensitive algorithms like K-Nearest Neighbors (KNN) or Support Vector Machines (SVM).

6 Machine Learning Algorithms

- **Suitable Models:** Decision Trees, Random Forests, Support Vector Machines (RBF/Linear kernel), K-Nearest Neighbors (KNN).
- **Feature Selection:** ANOVA F-test via `SelectKBest` or Principal Component Analysis (PCA) for dimensionality reduction.

7 Experimental Setup

Executed on Python 3.10+ using Pandas, NumPy, Matplotlib, and Seaborn on Windows 11.

8 Hyperparameter Selection

Density kernel bandwidths and pairwise bin sizing were determined dynamically using Seaborn defaults.

9 Results

All 17 visualization outputs were successfully generated and stored in `figures_iris/`.

10 Result Visualization

The visualizations demonstrate that petal dimensions (`PetalLengthCm` and `PetalWidthCm`) provide clean multi-class discrimination boundaries.

11 Discussion

The Iris dataset provides a foundational benchmark for multiclass linear vs. non-linear decision boundaries.

- **Strengths:** Small dimension, balanced classes, zero null values.
- **Insights:** Setosa requires only simple threshold rules, whereas Virginica and Versicolor require soft-margin classifiers.

12 Conclusion

The automated EDA framework effectively characterized the Iris dataset, establishing that petal length and width serve as optimal predictive attributes for species classification.

13 References

- [1] Fisher, R. A., “The use of multiple measurements in taxonomic problems,” *Annals of Eugenics*, 1936.
- [2] UCI Machine Learning Repository, “Iris Dataset,” <https://archive.ics.uci.edu/ml/datasets/iris>.